

My Literate Emacs Configuration

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1 Startup

1.1 Early initialization

The `early-init.el` removes all the visual fluff like scrollbars and sets up very fundamental Emacs settings that need to be loaded before everything else.

```

(setq inhibit-startup-message t)
(setq inhibit-startup-screen t)
(setq frame-resize-pixelwise t
      frame-inhibit-implied-resize t
      use-dialog-box t ; only for mouse events, which I seldom use
      use-file-dialog nil
      use-short-answers t
      inhibit-x-resources t
      inhibit-startup-echo-area-message user-login-name
      inhibit-startup-buffer-menu t)
(setq mode-line-misc-info
      (delete (assoc 'minor-mode-alist mode-line-misc-info)
              mode-line-misc-info))
(add-hook 'after-init-hook (lambda () (set-frame-name "home")))
;; Modes
(scroll-bar-mode -1)
(tool-bar-mode -1)
(tooltip-mode -1)
(set-fringe-mode 10)
(menu-bar-mode -1)
(tool-bar-mode 0)
;; settings for windows
(setq focus-follows-mouse t)
(add-to-list 'default-frame-alist '(alpha . (98 . 98)))
(add-to-list 'default-frame-alist '(undecorated . t))
(setq initial-frame-alist default-frame-alist)
;; garbage collection
(setq gc-cons-threshold most-positive-fixnum
      gc-cons-percentage 0.5)
(defvar bard-emacs--file-name-handler-alist file-name-handler-alist)
(defvar bard-emacs--vc-handled-backends vc-handled-backends)
(add-hook 'emacs-startup-hook
          (lambda ()
            (setq gc-cons-threshold (* 1000 1000 8)
                  gc-cons-percentage 0.1
                  file-name-handler-alist
                  bard-emacs--file-name-handler-alist
                  vc-handled-backends
                  bard-emacs--vc-handled-backends)))
(add-hook 'emacs-startup-hook

```



```
(lambda ()
  (profiler-stop)
  (profiler-report)))
```

1.2 Initialization

```
;;; init.el --- init.el -*- lexical-binding: t -*-
;; Author: BardofSprites
;; Maintainer: BardofSprites
;; Version: 0.1.0
;; Package-Requires: ((emacs "28.2"))
;; This file is not part of GNU Emacs
;; This program is free software: you can redistribute it and/or
;; modify
;; it under the terms of the GNU General Public License as published
;; by
;; the Free Software Foundation, either version 3 of the License, or
;; (at your option) any later version.
;; This program is distributed in the hope that it will be useful,
;; but WITHOUT ANY WARRANTY; without even the implied warranty of
;; MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
;; GNU General Public License for more details.
;; You should have received a copy of the GNU General Public License
;; along with this program. If not, see <https://www.gnu.org/licenses/>.
;;; Commentary:
;;; Code:
;; fix keymap
(repeat-mode t)
(global-set-key (kbd "C-z") nil)
(global-set-key (kbd "M-z") nil)
(global-set-key (kbd "C-z C-z") #'suspend-emacs)
;; Add the directories to the load path
(mapc
 (lambda (string)
   (add-to-list 'load-path (locate-user-emacs-file string)))
 '("bard-elisp" "bard-emacs-modules" "old-ada"))
;; Clipboard saving
(setq select-enable-clipboard t)
;; custom file is evil
(setq custom-file (make-temp-file "emacs-custom-"))
```

```

;; No Backups
(setq auto-save-default nil)
(setq make-backup-files nil)
(setq confirm-kill-emacs 'y-or-n-p)
;; native comp enabled
(when (native-comp-available-p)
  (setq native-compile-prune-cache t))
;; enable/disable commands
(mapc
 (lambda (command)
  (put command 'disabled nil))
 '(list-timers narrow-to-region narrow-to-page upcase-region
   downcase-region))
(mapc
 (lambda (command)
  (put command 'disabled t))
 '(eshell project-eshell overwrite-mode iconify-frame diary))
;;; Packages
(setq package-archives '(("elpa" . "https://elpa.gnu.org/packages/")
                        ("nongnu" . "https://elpa.nongnu.org/nongnu/"
                                   ("melpa" . "https://melpa.org/packages/"))))
;;; MACROS
(defmacro bard/make-abbrev (table &rest definitions)
  "Expand abbrev DEFINITIONS for the given TABLE.
DEFINITIONS is a sequence of (i) string pairs mapping the
abbreviation to its expansion or (ii) a string and symbol pair
making an abbreviation to a function."
  (declare (indent 1))
  (unless (zerop (% (length definitions) 2))
    (error "Uneven number of key+command pairs"))
  `(if (abbrev-table-p ,table)
      (progn
        ,@(mapcar
          (lambda (pair)
            (let ((abbrev (nth 0 pair))
                  (expansion (nth 1 pair)))
              (if (stringp expansion)
                  `(define-abbrev ,table ,abbrev ,expansion)
                  `(define-abbrev ,table ,abbrev "" ,expansion))))
          (seq-split definitions 2)))
      )

```

```

        (error "%s is not an abbrev table" ,table)))
(require 'bard-emacs-anki)
(require 'bard-emacs-calendar)
(require 'bard-emacs-completion)
(require 'bard-emacs-dired)
;; (require 'bard-emacs-email)
(require 'bard-emacs-media)
(require 'bard-emacs-eshell)
(require 'bard-emacs-essentials)
(require 'bard-emacs-modeline)
(require 'bard-emacs-org)
(require 'bard-emacs-prog)
(require 'bard-emacs-theme)
(require 'bard-emacs-ui)
(require 'bard-emacs-web)
(require 'bard-emacs-window)
(require 'bard-emacs-writing)
(provide 'init)
(put 'eshell 'disabled nil)

```

2 Modules

2.1 bard-emacs-anki

[Anki](#) is a flashcard program that helps you spend more time on challenging material, and less on what you already know.

This is an emacs package that allows me to write flashcards in org mode. I talked about it in my EmacsConf 2024 presentation titled: [A example of a cohesive student workflow in Emacs](#).

```

(use-package anki-editor
  :ensure t
  :after org
  :bind (:map org-mode-map
    ("C-c M-i" . bard/anki-editor-cloze-region-auto-incr)
    ("C-c M-I" . bard/anki-editor-cloze-region-dont-incr)
    ("C-c M-r" . bard/anki-editor-reset-cloze-number)
    ("C-c M-p" . anki-editor-push-new-notes)
    ("C-c M-P" . anki-editor-push-notes))
  :hook (org-capture-after-finalize . bard/

```

```

anki-editor-reset-cloze-number) ; Reset cloze-number after each
capture.
:config
(setq anki-editor-create-decks t ;; Allow anki-editor to create a
new deck if it doesn't exist
      anki-editor-org-tags-as-anki-tags t)
(defun bard/anki-editor-cloze-region-auto-incr (&optional arg)
  "Cloze region without hint and increase card number."
  (interactive)
  (anki-editor-cloze-region my-anki-editor-cloze-number "")
  (setq my-anki-editor-cloze-number (1+
my-anki-editor-cloze-number))
  (forward-sexp))
(defun bard/anki-editor-cloze-region-dont-incr (&optional arg)
  "Cloze region without hint using the previous card number."
  (interactive)
  (anki-editor-cloze-region (1- my-anki-editor-cloze-number) "")
  (forward-sexp))
(defun bard/anki-editor-reset-cloze-number (&optional arg)
  "Reset cloze number to ARG or 1"
  (interactive)
  (setq my-anki-editor-cloze-number (or arg 1)))
;; Initialize
(bard/anki-editor-reset-cloze-number)
)
(provide 'bard-emacs-anki)

```

2.2 bard-emacs-calendar

2.2.1 Load required libraries

```

(require 'bard-calendar)
(require 'hi-lock)

```

2.2.2 Org directory and agenda files

```

(setq org-directory "~/Notes/denote/")
;; symlinked file to shorten denote file name in agenda buffers
(setq org-agenda-files (list "~/Notes/denote/todo.org" "~/Notes/
denote/uni.org"))

```

2.2.3 Org Calendar

```
;; Calendar
(use-package calendar-mode
  :config
  (setq calendar-holidays (append calendar-holidays russian-holidays
    ))
  :hook
  (calendar-today-visible . calendar-mark-today)
  (calendar-mode . denote-journal-calendar-mode)
  (calendar-today-visible . calendar-mark-holidays))
```

1. Orthodox Christian Holidays

```
(use-package orthodox-christian-new-calendar-holidays
  :ensure t
  :config
  (setq holiday-other-holidays (append holiday-other-holidays
    orthodox-christian-new-calendar-holidays))
  (setq holiday-bahai-holidays nil
    holiday-christian-holidays nil
    holiday-islamic-holidays nil))
```

2.2.4 Org todo keywords

```
;; Org todo keywords - changed to using hl-todo faces fixed by modus
/ef themes
(setq org-todo-keywords
  '((sequence "TODO(t)" "EXTRA(e)" "INPROG(i)" "|" "DONE(d)" "
  KILLED(k)")
    (sequence "MEET(m)" "TENT(T)" "|" "MET(M)" "NOGO(n)")))
(setq org-todo-keyword-faces
  '(("EXTRA" . (:inherit warning))
    ("TENT" . (:inherit warning))
    ("MEET" . (:inherit warning underline bold))
    ("INPROG" . (:inherit hi-yellow :weight bold))))
(setq org-enforce-todo-dependencies t)
;; Automatically clock in
(add-hook 'org-after-todo-state-change-hook #'bard/auto-clock-in)
```

2.2.5 Org clocking

```
;; clock tables
(setq org-clock-clocktable-default-properties '(:maxlevel 7 :scope
agenda)
      org-clock-persist 'history
      org-clock-mode-line-total 'current)
(org-clock-persistence-insinuate)
(use-package org
  ;; not really show what this does anymore
  :demand nil
  :hook
  ((org-clock-out . bard/org-clock-update-mode-line)))
```

2.2.6 Org agenda settings

```
(global-set-key (kbd "<f1>") 'bard/default-agenda)
(global-set-key (kbd "M-<f1>") 'bard/choose-agenda)
(global-set-key (kbd "C-c a") 'org-agenda)
(setq org-agenda-include-diary t)
(setq org-agenda-custom-commands
  `(("D" "Daily agenda and top priority tasks"
      ((tags-todo "!TODO"
          ((org-agenda-overriding-header "Unscheduled
Tasks \n")
          (org-agenda-skip-function '(
org-agenda-skip-entry-if 'timestamp))))
      (agenda "" ((org-agenda-span 1)
                    (org-agenda-start-day nil)
                    (org-deadline-warning-days 14)
                    ;; (org-scheduled-past-days 0)
                    (org-agenda-day-face-function (lambda (date) '
org-agenda-date))
                    (org-agenda-format-date "%A %-e %B %Y")
                    (org-agenda-overriding-header "Today's agenda
\n"))))
      (agenda "" ((org-agenda-span 8)
                    (org-calendar-holiday)
                    (org-deadline-warning-days 0)
                    (org-agenda-skip-function '(
org-agenda-skip-entry-if 'todo 'done))
```

```

                                (org-agenda-overriding-header "Upcoming this
week \n")))))
  ("K" "Kanban board for projects"
    ((todo "TODO"
      ((org-agenda-files '("~/Notes/denote/kanban.org"))
       (org-agenda-overriding-header "Backlog")
       (org-agenda-sorting-strategy '(category-up))))
     (todo "INPROG"
      ((org-agenda-files '("~/Notes/denote/kanban.org"))
       (org-agenda-overriding-header "In Progress")
       (org-agenda-sorting-strategy '(category-up))))
     (todo "DONE"
      ((org-agenda-files '("~/Notes/denote/kanban.org"))
       (org-agenda-overriding-header "Completed projects")
       (org-agenda-sorting-strategy '(category-up))))))
  ("Y" "Yearly view for all tasks"
    ((agenda "" ((org-agenda-span 365)
                  (org-deadline-warning-days 2)
                  (org-agenda-skip-function '(
org-agenda-skip-entry-if 'todo 'done))
                  (org-agenda-overriding-header "Upcoming this
Year\n")))))
  ("M" "Monthly view for all tasks"
    ((agenda "" ((org-agenda-span 31)
                  (org-deadline-warning-days 2)
                  (org-agenda-skip-function '(
org-agenda-skip-entry-if 'todo 'done))
                  (org-agenda-overriding-header "Upcoming this
month\n"))))))))

```

2.2.7 Org habit

```

(use-package org-habit
  :after org-agenda
  :config
  (setq org-habit-show-done-always-green t
        org-habit-show-habits t
        org-habit-show-all-today t))

```

2.2.8 Provide module

```
(provide 'bard-emacs-calendar)
```

2.3 bard-emacs-completion

2.3.1 Vertico completion framework

```
;; minibuffer completion
(use-package vertico
  :ensure t
  :config
  (setq vertico-scroll-margin 0)
  (setq vertico-cycle nil)
  (setq vertico-resize nil)
  (vertico-mode t)
  (with-eval-after-load 'rfn-eshadow
    ;; This works with `file-name-shadow-mode' enabled.  When you
    are in
    ;; a sub-directory and use, say, `find-file' to go to your home
    '~/ '
    ;; or root '/' directory, Vertico will clear the old path to
    keep
    ;; only your current input.
    (add-hook 'rfn-eshadow-update-overlay-hook #'
      vertico-directory-tidy)))
```

2.3.2 Mct completion framework

```
(use-package mct
  :ensure t
  :defer t
  :config
  (setq mct-hide-completion-mode-line nil
        mct-live-completion t)
  (add-hook 'completion-list-mode-hook #'
    consult-preview-at-point-mode))
```

2.3.3 Rfn eshadow

```
(use-package rfn-eshadow
  :ensure nil
  :hook (minibuffer-setup . cursor-intangible-mode)
  :config)
```



```

(setq resize-mini-windows t)
(setq read-answer-short t) ; also check `use-short-answers' for
Emacs28
(setq echo-keystrokes 0.25)
;; Do not allow the cursor to move inside the minibuffer prompt.
I
;; got this from the documentation of Daniel Mendler's Vertico
;; package: <https://github.com/minad/vertico>.
(setq minibuffer-prompt-properties
  '(read-only t cursor-intangible t face minibuffer-prompt))
;; MCT has a variant of this built-in.
(defun crm-indicator (args)
  (cons (format "[`completing-read-multiple': %s]  %s"
              (propertize
                (replace-regexp-in-string
                  "\\`\\[\\.\\*?]\\|\\*\\|\\[\\.\\*?]\\|\\*\\|'" ""
                  crm-separator)
                'face 'error)
              (car args))
        (cdr args)))
(advice-add #'completing-read-multiple :filter-args #'
  crm-indicator)
(file-name-shadow-mode 1))

```

2.3.4 Marginalia

```

(use-package marginalia
  :ensure t
  :init
  (marginalia-mode 1))

```

2.3.5 Orderless

```

(use-package orderless
  :ensure t)

```

2.3.6 Tab completion

For a long time I really struggled with Emacs tab completion. It still only kind of works most of the time, but I don't want to tweak it to make it work. It works really well with LSP and using Lisp REPL's for languages like Clojure, Common Lisp, and Scheme. However for some

languages like Haskell, without the LSP (which hasn't worked for me in the past), it does not work to the extent that I believe it should. Completion is something I'm kind of jealous of when it comes to Neovim vs Emacs.

1. UI for tab completion

```
(use-package corfu
  :ensure t
  :init (global-corfu-mode)
  ;; I also have (setq tab-always-indent 'complete) for TAB to
  complete
  ;; when it does not need to perform an indentation change.
  :bind (:map corfu-map ("<tab>" . corfu-complete))
  :config
  (setq corfu-preview-current nil)
  (setq corfu-min-width 20)
  (setq corfu-popupinfo-delay nil)
  (corfu-popupinfo-mode 1) ; shows documentation after `
  corfu-popupinfo-delay'
  ;; Sort by input history (no need to modify `
  corfu-sort-function').
  (with-eval-after-load 'savehist
    (corfu-history-mode 1)
    (add-to-list 'savehist-additional-variables 'corfu-history))
  )
```

2. Compilation at point functions (CAPF)

```
(use-package cape
  :ensure t
  :init
  (add-to-list 'completion-at-point-functions #'cape-file)
  (add-to-list 'completion-at-point-functions #'cape-dabbrev)
  (add-to-list 'completion-at-point-functions #'cape-keyword)
  (add-to-list 'completion-at-point-functions #'cape-abbrev))
(use-package clang-capf
  :ensure t
  :config
  (defun bard/clang-capf-init ()
    "Add `clang-capf' to `completion-at-point-functions'."
    (add-hook 'completion-at-point-functions #'clang-capf nil t)
  )
```

```
(add-hook 'c-mode-hook #'bard/clang-capf-init))
```

3. Completion styles

```
(use-package minibuffer
  :config
  ;;;; Completion styles
  (setq completion-styles '(orderless basic substring initials
    flex))
  (setq completion-category-defaults nil)
  (setq completion-category-overrides
    '((file (styles . (orderless basic partial-completion)))
      (bookmark (styles . (basic substring)))
      (library (styles . (basic substring)))
      (embark-keybinding (styles . (basic substring)))
      (imenu (styles . (orderless basic substring)))
      (consult-location (styles . (orderless basic substring
    )))
      (kill-ring (styles . (emacs22 orderless)))
      (eglot (styles . (orderless emacs22 substring))))))
```

2.3.7 Minibuffer enhancements

1. Consult

```
(use-package consult
  :ensure t
  :defer 2
  :bind*
  ("C-x r b" . consult-bookmark)
  ("M-g M-g" . consult-goto-line)
  ("C-x b" . consult-buffer)
  ("M-s M-f" . consult-fd)
  ("M-s M-g" . consult-grep)
  ("M-s M-h" . consult-history)
  ("M-s M-y" . consult-yank-pop)
  ("M-s M-o" . consult-outline)
  ("M-s M-l" . consult-line)
  ("M-s M-k" . consult-kmacro)
  ("M-s M-r" . consult-register)
  :config
  (setq consult-find-args
```

```
(concat "find . -not ( "
      "-path */.git* -prune "
      "-or -path */.cache* -prune )")))
```

2. Embark

```
(use-package embark
  :ensure t
  :bind*
  (("C-," . bard-embark-act-no-quit)
   ("C-." . bard-embark-act-quit))
  :config
  (require 'bard-embark)
  (setq embark-keymap-alist
    '((buffer bard-embark-buffer-map)
      (command bard-embark-command-map)
      (expression bard-embark-expression-map)
      (file bard-embark-file-map)
      (function bard-embark-function-map)
      (identifier bard-embark-identifier-map)
      (package bard-embark-package-map)
      (region bard-embark-region-map)
      (symbol bard-embark-symbol-map)
      (url bard-embark-url-map)
      (variable bard-embark-variable-map)
      (t embark-general-map)))
  (defun bard-embark-act-no-quit ()
    "Call `embark-act' but do not quit after the action."
    (interactive)
    (let ((embark-quit-after-action nil))
      (call-interactively #'embark-act)))
  (defun bard-embark-act-quit ()
    "Call `embark-act' and quit after the action."
    (interactive)
    (let ((embark-quit-after-action t))
      (call-interactively #'embark-act))
    (when (and (> (minibuffer-depth) 0)
              (derived-mode-p 'completion-list-mode))
      (abort-recursive-edit)))
  (setq embark-confirm-act-all nil)
  ;; The prot-embark.el has an advice to further simplify the
  ;; minimal indicator. It shows cycling, which I never want to
```

```

    see
;; or do.
(setq embark-mixed-indicator-both nil)
(setq embark-mixed-indicator-delay 0.1)
(setq embark-indicators '(embark-mixed-indicator
  embark-highlight-indicator))
(setq embark-verbose-indicator-nested nil) ; I think I don't
  have them, but I do not want them either
(setq embark-verbose-indicator-buffer-sections '(bindings))
(setq embark-verbose-indicator-excluded-actions
  '(embark-cycle embark-act-all embark-collect
    embark-export embark-insert)))

```

2.3.8 Save minibuffer history (**savehist**)

```

(setq savehist-file (locate-user-emacs-file "savehist"))
(setq history-length 100)
(setq history-delete-duplicates t)
(setq savehist-save-minibuffer-history t)
(setq savehist-additional-variables '(register-alist kill-ring))
(savehist-mode t)

```

2.3.9 Recent files (**recentf**)

```

(recentf-mode t)
(setq recentf-save-file (locate-user-emacs-file "recentf"))
;; save cursor place for code files
(add-hook 'prog-mode-hook #'save-place-local-mode)

```

2.3.10 Abbreviations ((d)abbrev-mode)

```

(setq abbrev-file-name (locate-user-emacs-file "abbrevs"))
(setq only-global-abbrevs nil)
(bard/make-abbrev global-abbrev-table
  "meweb" "https://bardman.dev"
  "megit" "https://github.com/BardofSprites"
  "meyt" "https://www.youtube.com/@bardmandev"
  "protweb" "https://protesilaos.com/")
(bard/make-abbrev text-mode-abbrev-table
  "asciidoc" "AsciiDoc"
  "auctex" "AUCTeX")

```

"cafe"	"écaf"
"cliche"	"éclich"
"clojurescript"	"ClojureScript"
"emacsconf"	"EmacsConf"
"github"	"GitHub"
"gitlab"	"GitLab"
"javascript"	"JavaScript"
"latex"	"LaTeX"
"libreplanet"	"LibrePlanet"
"linkedin"	"LinkedIn"
"paypal"	"PayPal"
"sourcehut"	"SourceHut"
"texmacs"	"TeXmacs"
"typescript"	"TypeScript"
"visavis"	"àvis--vis"
"vscode"	"Visual Studio Code"
"youtube"	"YouTube"
"Результат"	"Результат= Сегодняшний Битвый="
"asf"	"and so on and so forth"
"paragraph"	"¶"
"em"	"_"
"ua"	"↑"
"da"	"↓"
"ra"	"→"
"la"	"←"
"xmonad"	"XMonad"
"xmobar"	"XMobar")

```

(dolist (hook '(text-mode-hook prog-mode-hook git-commit-mode-hook))
  (add-hook hook #'abbrev-mode))
(remove-hook 'save-some-buffers-functions #'abbrev--possibly-save)

```

2.3.11 Provide module

```

(provide 'bard-emacs-completion)
;;; bard-emacs-completion.el ends here

```

2.4 bard-emacs-dired

2.4.1 Dired subtree

```

(use-package dired-subtree
  :ensure t

```

```
:config
(setq dired-subtree-use-backgrounds nil))
```

2.4.2 Dired file preview

```
(use-package dired-preview
  :ensure t
  :bind
  (:map dired-mode-map
        ("P" . dired-preview-mode))
  :config
  (setq dired-preview-delay 0.1))
```

2.4.3 General dired settings

```
(use-package dired
  :bind*
  (("C-j" . dired-jump))
  :bind (:map dired-mode-map
              ("E" . emms-add-dired)
             ("<tab>" . dired-subtree-toggle)
              ("<backtab>" . dired-subtree-cycle)))

:config
(setq dired-guess-shell-alist-user ; those are the suggestions for
! and & in Dired
  '(("\\.\\.(png\\|jpe?g\\|tiff\\)" "nsxiv" "feh" "xdg-open")
    ("\\.\\.(mp[34]\\|m4a\\|ogg\\|flac\\|webm\\|mkv\\)" "mpv" "
xdg-open")
    (".gif" "mpv --loop=inf")
    (".*" "xdg-open")))
(setq dired-dwim-target t)
(setq dired-listing-switches
  "-AgGFhlv --group-directories-first --time-style=long-iso")
:hook
((dired-mode . dired-hide-details-mode)
 ;; attachments for email through dired
 (dired-mode . turn-on-gnus-dired-mode)))
```

2.4.4 Image dired

```
(use-package image-dired
```

```

:bind
(:map dired-mode-map
      (("") . image-dired-dired-display-external)
      ("B" . bard/dired-set-background-with-feh))
:bind
("C-x C-d" . image-dired)
:bind
(:map image-dired-thumbnail-mode-map ("B" . bard/
    image-dired-set-background-with-feh))
:config
(define-advice image-dired-display-image (:override (file &
    optional _ignored))
  (setq file (expand-file-name file))
  (when (not (file-exists-p file))
    (error "No such file: %s" file))
  (let ((buf (get-buffer image-dired-display-image-buffer))
        (cur-win (selected-window)))
    (when buf
      (kill-buffer buf))
    (when-let ((buf (find-file-noselect file nil t)))
      (with-current-buffer buf
        (rename-buffer image-dired-display-image-buffer)
        (if (string-match (image-file-name-regexp) file)
            (image-dired-image-mode)
            ;; Support visiting PDF files.
            (normal-mode))
        (display-buffer buf))
      (select-window cur-win))))
(setq image-dired-thumbnail-storage 'standard)
(setq image-dired-external-viewer "nsxiv")
(setq image-dired-thumb-size 80)
(setq image-dired-thumb-margin 2)
(setq image-dired-thumb-relief 0)
(setq image-dired-thumbs-per-row 4)
(defun bard/dired-set-background-with-feh ()
  "Set the selected image as the background using feh."
  (interactive)
  (let ((image-file (dired-get-file-for-visit)))
    (start-process "feh" nil "feh" "--bg-fill" image-file)
    (message "Background set to %s" image-file)))
(defun bard/image-dired-set-background-with-feh ()

```



```
"Set the selected image as the background using feh."
(interactive)
(let ((image-file (image-dired-original-file-name)))
  (start-process "feh" nil "feh" "--bg-fill" image-file)
  (message "Background set to %s" image-file))))
```

2.4.5 Running commands on dired files

```
;; Taken from https://superuser.com/a/176629
(defun bard/dired-do-command (command)
  "Run COMMAND on marked files. Any files not already open will be
  opened.
  After this command has been run, any buffers it's modified will
  remain
  open and unsaved."
  (interactive "CRun on marked files M-x ")
  (save-window-excursion
    (mapc (lambda (filename)
            (find-file filename)
            (call-interactively command))
          (dired-get-marked-files)))))
```

2.4.6 Dired video thumbnails

```
(use-package dired-video-thumbnail
  :ensure t
  :vc (:url "https://github.com/captainflasmr/dired-video-thumbnail"
        :rev :newest)
  :bind (:map dired-mode-map
            ("C-t v" . dired-video-thumbnail)))
```

2.4.7 Provide module

```
(provide 'bard-emacs-dired)
```

2.5 bard-emacs-email

2.5.1 Require email library

```
(require 'bard-email)
```

2.5.2 Notmuch mail client settings

```
(use-package notmuch
  :ensure t
  :config
  (define-key global-map (kbd "C-c m") #'notmuch)
  (setq notmuch-show-logo t
        notmuch-column-control 1.0
        notmuch-hello-auto-refresh t
        notmuch-hello-recent-searches-max 20
        notmuch-hello-thousands-separator ""
        notmuch-hello-sections '(notmuch-hello-insert-header
                                   notmuch-hello-insert-saved-searches notmuch-hello-insert-search
                                   notmuch-hello-insert-alltags)
        notmuch-show-all-tags-list t)
  (setq notmuch-search-oldest-first nil)
  (setq notmuch-show-seen-current-message t)
  (defun bard/notmuch-mua-empty-subject-check ()
    "Request confirmation before sending a message with empty
    subject."
    (when (and (null (message-field-value "Subject"))
               (not (y-or-n-p "Subject is empty, send anyway? ")))
      (error "Sending message cancelled: empty subject")))
  (add-hook 'message-send-hook 'bard/notmuch-mua-empty-subject-check
    ))
```

1. Notmuch saved searches

```
(setq notmuch-show-empty-saved-searches t)
(setq notmuch-saved-searches
  `(( :name "☐ inbox (all-mail)"
      :query "tag:inbox"
      :sort-order newest-first
      :key ,(kbd "i"))
    ( :name "☐ unread (inbox)"
      :query "tag:unread and tag:inbox"
      :sort-order newest-first
      :key ,(kbd "u"))
    ( :name "☐ To Do"
      :query "tag:todo"
      :sort-order oldest-first
      :key ,(kbd "t"))
```

```
( :name "␣ flagged"
  :query "tag:flag"
  :sort-order newest-first
  :key ,(kbd "f"))
( :name "␣ contributions"
  :query "tag:unread and tag:contrib"
  :sort-order newest-first
  :key ,(kbd "c"))
( :name "␣ linux-related"
  :query "tag:unread and tag:linux"
  :sort-order newest-first
  :key ,(kbd "l"))
( :name "␣ emacs development"
  :query "tag:unread and tag:contrib"
  :sort-order newest-first
  :key ,(kbd "ed"))
( :name "␣ emacs humanities"
  :query "tag:unread and tag:emacs-humanities"
  :sort-order newest-first
  :key ,(kbd "eh"))
( :name "␣ emacs org-mode"
  :query "tag:unread and tag:emacs-org"
  :sort-order newest-first
  :key ,(kbd "eo"))))
```

2. Tagging keys

```
(setq notmuch-tagging-keys
  `((,(kbd "d") prot-notmuch-mark-delete-tags "␣ Mark for
    deletion")
    ,(kbd "f") prot-notmuch-mark-flag-tags "␣ Flag as
    important")
    ,(kbd "s") prot-notmuch-mark-spam-tags "␣ Mark as spam"
  )
  ,(kbd "r") ("unread") "␣␣ Mark as read")
  ,(kbd "u") ("unread") "␣ Mark as unread"))
(setq notmuch-archive-tags '("+archive")
  notmuch-message-replied-tags '("+replied")
  notmuch-message-forwarded-tags '("+forwarded")
  notmuch-show-mark-read-tags '("-unread")
  notmuch-draft-tags '("+draft")
  notmuch-draft-folder "drafts")
```

```
notmuch-draft-save-plaintext 'ask)
```

2.5.3 Notmuch modeline indicator

```
(use-package notmuch-indicator
  :ensure t
  :after notmuch
  :config
  (setq notmuch-indicator-args
    '(( :terms "tag:unread and tag:inbox"
        :label "[U] "
        :label-face prot-modeline-indicator-green
        :counter-face prot-modeline-indicator-green)
      ( :terms "tag:unread and tag:linux"
        :label "[L] "
        :label-face prot-modeline-indicator-cyan
        :counter-face prot-modeline-indicator-cyan)
      ( :terms "tag:unread and tag:emacs"
        :label "[E] "
        :label-face prot-modeline-indicator-blue
        :counter-face prot-modeline-indicator-blue))
    notmuch-indicator-refresh-count (* 60 3)
    notmuch-indicator-hide-empty-counters t
    notmuch-indicator-force-refresh-commands '(
      notmuch-refresh-this-buffer))
  (setq notmuch-indicator-add-to-mode-line-misc-info nil)
  (notmuch-indicator-mode t))
```

2.5.4 Org links for notmuch

```
(use-package ol-notmuch
  :ensure t
  :after notmuch)
```

2.5.5 Send mail and smtp settings

```
;; use msmtplib
(setq sendmail-program "/usr/bin/msmtplib"
  message-send-mail-function 'message-send-mail-with-sendmail
  message-sendmail-f-is-evil nil
  mail-specify-envelope-from t)
```

```
message-sendmail-envelope-from 'header
mail-envelope-from t)
```

2.5.6 Provide module

```
(provide 'bard-emacs-email)
```

2.6 bard-emacs-eshell

```
(use-package eshell
  :ensure nil
  :bind
  (("C-z e" . eshell-switcher))
  :config
  (require 'bard-eshell)
  (require 'ffap)
  ;; (setq eshell-banner-message "Time for another recreational
    programming session.\n\n")
  (setq eshell-banner-message
    '(format "%s %s\n %s\n"
              (propertize (format " %s " (string-trim (
buffer-name)))
                          'face 'mode-line-highlight)
              (propertize (current-time-string)
                          'face 'font-lock-keyword-face)
              (propertize "Time for another recreational programming
session."
                          'face 'warning))))
  (setq bard/eshell-aliases
    '((g . magit)
      (gl . magit-log)
      (d . dired)
      (o . find-file)
      (oo . find-file-other-window)
      (vim . find-file)
      (l . (lambda () (eshell/ls '-la)))
      (eshell/clear . eshell/clear-scrollback)))
  (mapc (lambda (alias)
          (defalias (car alias) (cdr alias)))
        bard/eshell-aliases))
(use-package eshell
```

```

:ensure nil
:after esh-mode
:bind
(:map eshell-mode-map
  ("C-c C-e" . prot-eshell-export)
  ("M-k" . eshell-kill-input)
  ("C-c C-d" . prot-eshell-complete-recent-dir)
  ("C-c C-h" . prot-eshell-narrow-output-highlight-regexp)
  ("C-c C-f" . bard/eshell-find-file-at-point)))
(provide 'bard-emacs-eshell)

```

2.7 bard-emacs-essentials

2.7.1 Grep

1. Settings for Emacs grep

```

(use-package grep
  :ensure t
  :config
  (setq grep-use-headings t))

```

2. Writable grep buffers (wgrep)

```

(use-package wgrep
  :ensure t
  :bind
  (:map wgrep-mode-map
    ("C-x C-s" . wgrep-save-all-buffers)
    ("C-x C-q" . wgrep-change-to-wgrep-mode)
    ("C-c C-c" . wgrep-finish-edit))
  :bind
  (:map grep-mode-map
    ("e" . wgrep-change-to-wgrep-mode)))

```

2.7.2 Multiple cursors

```

(use-package multiple-cursors
  :ensure t
  :config
  (setq mc/always-run-for-all t)
  :bind

```

```
(("C-S-c C-S-c" . mc/edit-lines)
 ("C->" . mc/mark-next-like-this)
 ("C-<" . mc/mark-previous-like-this)
 ("C-c C" . mc/mark-all-like-this)
 ("C-\" . mc/skip-to-next-like-this)
 ("C-;" . mc/skip-to-previous-like-this)))
```

2.7.3 Expand region

```
(use-package expand-region
 :ensure t
 :bind ("C-=" . er/expand-region))
```

2.7.4 Substitute (simple text substitution)

```
(use-package substitute
 :ensure t
 :bind
 (("C-c s b" . substitute-target-below-point)
 ("C-c s a" . substitute-target-above-point)
 ("C-c s d" . substitute-target-in-defun)
 ("C-c s s" . substitute-target-in-buffer)))
```

2.7.5 Saving session with desktop

```
;; Desktop mode/session saving
(setq desktop-path '("~/emacs.d/desktop")
      desktop-dirname "~/emacs.d/desktop/"
      desktop-base-file-name "emacs-desktop"
      desktop-save t
      desktop-restore-eager t
      desktop-restore-frames t
      desktop-restore-in-current-display t
      desktop-files-not-to-save "(^$\\|\\*scratch\\*\\|\\*Messages
\\*\\|\\*dashboard\\*\\|\\*Async-native-compile-log\\*\\|\\*Music
\\*)" )
;; (desktop-save-mode t)
(global-set-key (kbd "C-z s") 'desktop-save-in-desktop-dir)
(global-set-key (kbd "C-z r") 'desktop-read)
```

2.7.6 Scratch buffers

```
;; Text Scratch buffers
(defun bard/new-org-buffer ()
  (interactive)
  (let ((xbuf (generate-new-buffer "*org*")))
    (switch-to-buffer xbuf)
    (funcall (quote org-mode))
    (text-scale-increase 1.5)
    xbuf))
(define-key global-map (kbd "M-=") #'bard/new-org-buffer)
(defun bard/new-plain-buffer ()
  (interactive)
  (let ((xbuf (generate-new-buffer "*plain*")))
    (switch-to-buffer xbuf)
    (text-scale-increase 1.5)
    xbuf))
(define-key global-map (kbd "M--") #'bard/new-plain-buffer)
;; elisp scratch buffer
(defun bard/new-elisp-buffer ()
  (interactive)
  (let ((xbuf (generate-new-buffer "*elisp*")))
    (switch-to-buffer xbuf)
    (funcall (quote emacs-lisp-mode))
    (text-scale-increase 1.5)
    xbuf))
```

2.7.7 Terminals

```
(defun bard/open-terminal-in-current-directory ()
  "Open a terminal in the current working directory."
  (interactive)
  (let ((default-directory default-directory))
    (term "/usr/bin/env bash")))
;; (define-key global-map (kbd "C-t") #'bard/
  open-terminal-in-current-directory)
(define-key global-map (kbd "C-z t") #'bard/
  open-terminal-in-current-directory)
(defun bard/open-terminal-emulator ()
  "Open a terminal in the current working directory."
  (interactive)
  (let ((default-directory default-directory))
```



```
(start-process "st terminal" nil "st"))
(define-key global-map (kbd "C-z C-t") 'bard/open-terminal-emulator)
(define-key global-map (kbd "C-z C-s") #'bard/new-elisp-buffer)
```

2.7.8 Time Management

```
;; Modeline
(setq display-time-format "%Y-%m-%d (%a) %H:%M")
(setq display-time-interval 60)
(setq display-time-default-load-average nil)
(setq display-time-mail-directory nil)
(setq display-time-mail-function nil)
(setq display-time-use-mail-icon nil)
(setq display-time-mail-string nil)
(setq display-time-mail-face nil)
(setq display-time-string-forms
  '((propertize
    (format-time-string display-time-format now)
    'face 'display-time-date-and-time
    'help-echo (format-time-string "%a %b %e, %Y" now))
    " "))
(display-time-mode 1)
;; world clock
(setq world-clock-list
  '(("America/New_York" "New York")
    ("Europe/Moscow" "Moscow")
    ("Europe/London" "London")
    ("Asia/Tokyo" "Tokyo")))
(setq world-clock-time-format "%Y-%m-%d %B (%A) %R %Z")
```

2.7.9 Timer (tmr)

```
;; timer package
(use-package tmr
  :ensure t
  :config
  (setq tmr-notification-urgency 'normal)
  (setq tmr-sound-file nil)
  (setq tmr-timer-finished-functions '(tmr-notification-notify

tmr-print-message-for-finished-timer
```

```

                                tmr-acknowledge-minibuffer))
(setq tmr-descriptions-list 'tmr-description-history)
(define-key global-map (kbd "C-c t l") 'tmr-tabulated-view)
(define-key global-map (kbd "C-c t t") #'tmr)
(define-key global-map (kbd "C-c t T") #'tmr-with-details)
(define-key global-map (kbd "C-c t l") #'tmr-tabulated-view)
(define-key global-map (kbd "C-c t c") #'tmr-clone)
(define-key global-map (kbd "C-c t k") #'tmr-cancel)
(define-key global-map (kbd "C-c t s") #'tmr-reschedule)
(define-key global-map (kbd "C-c t e") #'tmr-edit-description)
(define-key global-map (kbd "C-c t r") #'tmr-remove)
(define-key global-map (kbd "C-c t R") #'tmr-remove-finished))

```

2.7.10 Bookmarks

```

(use-package bookmarks
  :ensure nil
  :hook
  (bookmark-bmenu-mode-hook . hl-line-mode)
  :config
  (setq bookmark-fringe-mark nil)
  (setq bookmark-save-flag 1)
  :bind ; C-c m for (book)-*m*ark
  ("C-c m" . bookmark-bmenu-list))

```

2.7.11 Running emacs as server

```

(require 'server)
(setq server-client-instructions nil)
(unless (server-running-p)
  (server-start))

```

2.7.12 Recording mode

```

(defun bard/recording-mode ()
  "A minor mode for recording Emacs videos."
  )

```

2.7.13 Provide module

```

(provide 'bard-emacs-essentials)

```

2.8 bard-emacs-media

2.8.1 EMMS music player

```
(use-package emms
  :ensure t
  :defer t
  :bind
  (:map emms-playlist-mode-map
    ("A" . emms-add-directory)
    ("l" . emms-add-playlist)
    ("T" . emms-add-directory-tree)
    ("F" . emms-add-file)
    ("U" . emms-add-url)
    ("L" . emms-toggle-repeat-track)
    ("<mouse-3>" . emms-pause)
    ("<SPC>" . emms-pause)
    ("c" . bard/emms-recenter)
    ("P" . emms-playlist-mode-shift-track-up)
    ("N" . emms-playlist-mode-shift-track-down)
    ("Z" . bard/save-emms-watch-later)
    ("Y" . bard/emms-download-current-video))
  :bind (("<f8>" . emms)
    ("M-<f8>" . emms-browser))
  :hook
  (emms-playlist-mode . hl-line-mode)
  :config
  (emms-all)
  (emms-default-players)
  (emms-mpris-enable)
  (setq emms-player-list '(emms-player-mpv))
  ;; emms-info-functions '(emms-info-native)
  ;; (setq emms-browser-covers 'emms-browser-cache-thumbnail)
  (setq emms-volume-amixer-card 0)
  ;; center line function
  (defun bard/emms-recenter ()
    (interactive)
    (recenter-top-bottom)
    (emms-playlist-mode-center-current))
  ;; modeline
  (emms-mode-line-disable)
  (emms-playing-time-disable-display))
```

```
;; playlist saving
(setq bard/emms-playlist-format 'm3u)
(setq bard/watch-later-file "~/Videos/watch-later.m3u")
)
```

2.8.2 YouTube browser (yeetube)

```
(use-package yeetube
  :ensure t
  :defer 30)
```

2.8.3 PDF viewer (pdf-tools)

```
(use-package pdf-tools
  :ensure t
  :config
  (pdf-tools-install)
  (add-to-list 'pdf-tools-enabled-modes #'pdf-view-themed-minor-mode)
  (add-to-list 'pdf-tools-enabled-modes #'pdf-view-roll-minor-mode)
  (defun bard/open-pdf-externally ()
    "Open current pdf file in Zathura."
    (interactive)
    (start-process "zathura" nil "zathura"
      "--page" (number-to-string (pdf-view-current-page
        (buffer-file-name)))
      (buffer-file-name)))
  :bind
  (:map pdf-view-mode-map
    ("&" . bard/open-pdf-externally)))
```

2.8.4 Epub reader (nov)

I use this package in tangent with [org-remark](#) to take notes on books as I am reading them.

```
(use-package nov
  :ensure t
  :after org-remark
  :config
  (add-to-list 'auto-mode-alist '("\\.epub\\\\" . nov-mode))
  :hook
  (nov-mode . logos-focus-mode))
```

```

:bind
(:map nov-mode-map
      ("a" . org-remark-mark)
      ("o" . org-remark-open)))

```

2.8.5 Custom functions

```

(use-package bard-media
  :ensure nil
  :config
  (require 'bard-media)
  :bind
  (("C-c o p" . bard/play-youtube-video)
   ("C-c o i" . bard/image-browser))
)

```

2.8.6 Provide module

```

(provide 'bard-emacs-media)

```

2.9 bard-emacs-modeline

```

(require 'bard-modeline)
;;; Mode line
(setq mode-line-compact nil) ; Emacs 28
(setq mode-line-right-align-edge 'right-margin)
(setq prot-modeline-show-frame-name (alist-get 'undecorated
  initial-frame-alist))
(setq-default mode-line-format
  '("%e"
    prot-modeline-kbd-macro
    prot-modeline-narrow
    bard-modeline-centered-cursor
    prot-modeline-input-method
    prot-modeline-buffer-status
    prot-modeline-window-dedicated-status
    bard-evil-state-indicator
    " "
    prot-modeline-buffer-identification
    " "
    prot-modeline-frame-name

```

```

        " "
        prot-modeline-major-mode
        prot-modeline-process
        " "
        prot-modeline-vc-branch
        " "
        prot-modeline-flymake
        prot-modeline-eglot
        bard-modeline-org-clock
        " "
        mode-line-format-right-align
        prot-modeline-notmuch-indicator
        " "
        prot-modeline-misc-info))
(with-eval-after-load 'spacious-padding
  (defun prot/modeline-spacious-indicators ()
    "Set box attribute to `prot-modeline-indicator-button' if
    spacious-padding is enabled."
    (if (bound-and-true-p spacious-padding-mode)
        (set-face-attribute 'prot-modeline-indicator-button nil :box
                             t)
        (set-face-attribute 'prot-modeline-indicator-button nil :box '
                             unspecified)))
;; Run it at startup and then afterwards whenever
;; `spacious-padding-mode' is toggled on/off.
(prot/modeline-spacious-indicators)
(add-hook 'spacious-padding-mode-hook #'prot/
          modeline-spacious-indicators))
(setq mode-line-right-align-edge 'window)
(provide 'bard-emacs-modeline)
;;; bard-emacs-modeline.el ends here

```

2.10 bard-emacs-org

2.10.1 Main Configuration

```

(use-package org
  :defer t
  :bind
  (:map org-mode-map
        ("C-M-a" . backward-paragraph)

```

```

("C-M-e" . forward-paragraph)
("C-c M-c" . count-words-region)
("C-c C-M-c" . count-words)
("C-c l" . org-id-get-create)
("C-c j" . org-goto)
)
:bind
(("C-c c" . org-capture))
:config
(setq org-goto-interface 'outline-path-completion
      org-outline-path-complete-in-steps nil
      org-goto-max-level '2)
(setq org-id-track-globally nil)
(setq org-id-search-archives nil)
(setq org-tags-column 0)
(setq org-special-ctrl-a/e t)
(setq safe-local-variable-values '((org-refile-targets (nil :
maxlevel . 3)))))

```

```

(setq org-archive-location "~/Notes/denote/20240328
T215840--archive__self.org::* Archive")
(setq org-log-done 'time)
(setq org-icalendar-include-todo t
      org-icalendar-include-body t
      org-icalendar-with-timestamps t
      org-icalendar-use-scheduled '(event-if-todo-not-done)
      org-icalendar-scheduled-summary-prefix "SCHEDULED: "
      org-icalendar-use-deadline '(event-if-todo-not-done)
      org-icalendar-deadline-summary-prefix "DEADLINE: ")

```

```

(setq org-structure-template-alist
      '(("c" . "center")
        ("x" . "eg")
        ("X" . "example")
        ("d" . "definition")
        ("t" . "theorem")
        ("n" . "NB")
        ("q" . "quote")
        ("v" . "verse")
        ("s" . "src")
        ("E" . "src emacs-lisp :results value code :lexical t") ;

```

```

for code examples in notes
  ("z" . "src emacs-lisp :tangle FILENAME") ; tangle without
making dir, below makes dir
  ("Z" . "src emacs-lisp :tangle FILENAME :mkdirp yes"))
(setq org-ellipsis " ⌘")

```

2.10.2 Encrypting org entries

```

(use-package org-crypt
  :config
  (setq org-crypt-key "develop.bard@gmail.com")
  (org-crypt-use-before-save-magic))

```

2.10.3 Making org mode look nice

```

(setq org-startup-indented t
      org-startup-folded 'showeverything
      org-hide-emphasis-markers t
      org-startup-with-inline-images t
      org-image-actual-width '(600)
      org-list-allow-alphabetical t
      org-insert-heading-respect-content t)
;; (use-package org-bullets
;;   :ensure t
;;   :hook (org-mode . org-bullets-mode)
;;   :config
;;   (setq org-bullets-bullet-list '("○" "●" "⊠" "⊡" "◆")))

```

2.10.4 Org export + latex

```

(setq org-format-latex-options (plist-put org-format-latex-options :
  scale 1.5))
(defun bard/org-export-on-save ()
  "Export current Org buffer to PDF and open it with auto-revert
  enabled."
  (when (derived-mode-p 'org-mode)
    (org-latex-export-to-pdf)))
(define-minor-mode bard/org-auto-export-pdf-mode
  "Automatically export Org buffer to PDF on save."
  :lighter " AutoPDF"

```



```

:group 'org
(if bard/org-auto-export-pdf-mode
  (add-hook 'after-save-hook #'bard/org-export-on-save)
  (remove-hook 'after-save-hook #'bard/org-export-on-save)))
(use-package auctex
  :ensure t)
;; latex editing niceness
(use-package org-fragtog
  :ensure t)
(with-eval-after-load 'ox-latex
  (add-to-list 'org-latex-classes
    '("org-plain-latex"
      "\\documentclass{article}
      [NO-DEFAULT-PACKAGES]
      [PACKAGES]
      [EXTRA]"
      ("\\section{%s}" . "\\section*{%s}")
      ("\\subsection{%s}" . "\\subsection*{%s}")
      ("\\subsubsection{%s}" . "\\subsubsection*{%s}")
      ("\\paragraph{%s}" . "\\paragraph*{%s}")
      ("\\subparagraph{%s}" . "\\subparagraph*{%s}")))))
(setq org-latex-listings t)
(setq org-latex-listings-options
  '(("basicstyle" "\\ttfamily")
    ("breakatwhitespace" "false")
    ("breakautoindent" "true")
    ("breaklines" "true")
    ("columns" "[c]fullflexible")
    ("commentstyle" "")
    ("emptylines" "*")
    ("extendedchars" "false")
    ("fancyvrb" "true")
    ("firstnumber" "auto")
    ("flexiblecolumns" "false")
    ("frame" "single")
    ("framaround" "tttt")
    ("identifierstyle" "")
    ("keepspaces" "true")
    ("keywordstyle" "")
    ("mathescape" "false")
    ("numbers" "left")

```

```

("numbers" "none")
("numbersep" "5pt")
("numberstyle" "\\tiny")
("resetmargins" "false")
("showlines" "true")
("showspaces" "false")
("showstringspaces" "false")
("showtabs" "true")
("stepnumber" "2")
("stringstyle" "")
("tab" "␣")
("tabsize" "4")
("texcl" "false")
("upquote" "false")))

```

2.10.5 Org capture

```

(setq org-capture-bookmark nil
  org-id-link-to-org-use-id 'create-if-interactive)
;; (require 'org-protocol)
(setq org-capture-templates
  '(("t" "task" entry
    (file+olp
      "~/Notes/denote/todo.org"
      "Inbox" "General tasks")
    "* TODO %?")
    ("a" "appointment" entry
      (file+olp
        "~/Notes/denote/todo.org"
        "Appointments" "General")
      "* %?\n%^t"
      :time-prompt t)
    ("d" "devlog entry" entry
      (file+olp+datetree "~/Notes/denote/20260511
T105305--development-log__programming.org")
      "* %u - %^{Project}\n%?"
      :tree-type month)
    ("s" "self learning task" plain
      (file+olp
        "~/Notes/denote/20251202
T154611--self-learning-agenda__topic.org"

```

```

    "Inbox")
    "- [ ] %?"
    :empty-lines-after 1)
    ("n" "common place note" entry
     (file "~/Notes/denote/20251023
T182240--common-place-notes__topic.org")
     "* %^{Source}\n#+BEGIN_QUOTE\n%\n#+END_QUOTE")
    ("z" "Protocol" entry
     (file+olp
      "~/Notes/denote/20240328
T220037--media-tracker__media_topic.org" "Quotes")
     "* Source: [%:link][%:description]]\n#+BEGIN_QUOTE\n%i\n#+
END_QUOTE\n%?"")
    ("Z" "Protocol Link" entry
     (file+olp
      "~/Notes/denote/20240328
T220037--media-tracker__media_topic.org" "Watch/Read List")
     "* [[%:link][%:description]] \nCaptured On: %U \n%?"")
    ("w" "Class outline" entry
     (file
      "~/Notes/denote/uni.org")
     (file
      "~/Notes/denote/templates/class-template.org"))
    ("p" "project idea" entry
     (file
      "~/Notes/denote/20250201
T165619--project-ideas__idea_programming.org")
     "* %^{Project description}\n%?""))

```

2.10.6 Managing media

```

;; copy/paste images
(use-package org-download
  :after org
  :ensure t
  :custom
  (org-download-method 'directory)
  (org-download-image-dir "~/Notes/denote/Images")
  (org-download-heading-lvl 0)
  (org-download-timestamp "org_%Y%m%d-%H%M%S_")
  (org-download-screenshot-method "wpaste -t image/png > '%s'"))

```

```

:bind
("C-M-y" . org-download-screenshot))
;; pdf notes
(use-package org-noter
  :ensure t
  :commands (org-noter))
(use-package org-pdftools
  :ensure t
  :after org-noter
  :defer t
  :commands (org-pdftools-setup-link)
  :hook (org-mode . org-pdftools-setup-link))
(use-package org-noter-pdftools
  :ensure t
  :after org-noter
  :defer t
  :config
  ;; Add a function to ensure precise note is inserted
  (defun org-noter-pdftools-insert-precise-note (&optional
    toggle-no-questions)
    (interactive "P")
    (org-noter--with-valid-session
      (let ((org-noter-insert-note-no-questions (if
        toggle-no-questions
                                          (not
org-noter-insert-note-no-questions)

org-noter-insert-note-no-questions)))
        (org-pdftools-use-isearch-link t)
        (org-pdftools-use-freepointer-annot t))
      (org-noter-insert-note (org-noter--get-precise-info))))))
;; fix https://github.com/weirdNox/org-noter/pull/93/commits/
f8349ae7575e599f375de1be6be2d0d5de4e6cbf
(defun org-noter-set-start-location (&optional arg)
  "When opening a session with this document, go to the current
location.
With a prefix ARG, remove start location."
  (interactive "P")
  (org-noter--with-valid-session
    (let ((inhibit-read-only t)
          (ast (org-noter--parse-root))

```

```

        (location (org-noter--doc-approx-location (when (
called-interactively-p 'any) 'interactive))))
        (with-current-buffer (org-noter--session-notes-buffer session
)
        (org-with-wide-buffer
        (goto-char (org-element-property :begin ast))
        (if arg
        (org-entry-delete nil org-noter-property-note-location
)
        (org-entry-put nil org-noter-property-note-location
        (org-noter--pretty-print-location
location)))))))))
(with-eval-after-load 'pdf-annot
  (add-hook 'pdf-annot-activate-handler-functions #'
    org-noter-pdf-tools-jump-to-note)))
;; links
(use-package org-cliplink
  :ensure t
  :bind
  ("C-c p" . org-cliplink))
(provide 'bard-emacs-org)

```

2.11 bard-emacs-prog

2.11.1 General input setup

```

(use-package electric
  :hook
  (prog-mode . electric-indent-local-mode)
  (prog-mode . electric-pair-local-mode))
(use-package paren
  :hook (prog-mode . show-paren-local-mode)
  :config
  (setq show-paren-style 'parenthesis)
  (setq show-paren-when-point-in-periphery nil)
  (setq show-paren-when-point-inside-paren nil)
  (setq show-paren-context-when-offscreen 'overlay))

```

2.11.2 Haskell

```

(use-package haskell-mode
  :ensure t

```

```
:defer t
:config
(setq haskell-interactive-popup-errors nil))
```

2.11.3 C/C++

```
(use-package c-mode
  :config
  (setq-default c-basic-offset 4)
  (setq c-default-style '((c-mode . "gnu")
                          (java-mode . "java")
                          (awk-mode . "awk"))))

(use-package compile
  :ensure nil
  :defer 2
  :config
  (require 'bard-compile)
  (setq compilation-scroll-output t
        compilation-auto-jump-to-first-error nil))
```

2.11.4 Lisp development

```
(use-package clojure-mode
  :ensure t
  :defer t)
(use-package cider
  :ensure t
  :defer t)
(use-package sly
  :ensure t
  :defer t
  :config
  (setq inferior-lisp-program (executable-find "sbcl")))
(use-package geiser
  :ensure t
  :defer t)
(use-package geiser-gauche
  :ensure t
  :defer t)
;; parens packages
```

2.11.5 Version control

```
;; Version control
(setq vc-follow-symlinks t)
(use-package magit
  :ensure t
  :config
  (setq magit-repository-directories
    '("~/Code" . 1)
      ("~/Repositories" . 1)
      ("~/dotfiles-stow" . 0)
      ("~/ .emacs.d" . 0)
      ("~/Pictures/wallpaper" . 0)))
  :bind ("C-c g" . magit-status))
```

2.11.6 Ada-mode

```
;; (use-package ada-mode
;;   :load-path "~/ .emacs.d/old-ada"
;;   :bind
;;   (:map ada-mode-map
;;     ["C-j" . dired-jump]))
```

2.11.7 Provide module

```
(provide 'bard-emacs-prog)
```

2.12 bard-emacs-theme

2.12.1 Load required libraries

```
(require 'bard-theme)
```

2.12.2 Set all themes to safe

```
;; declare all themes as safe (i trust developers)
(setq custom-safe-themes t)
```

2.12.3 Color theme

```
(use-package modus-themes
  :ensure t
```

```
:config
(bard/select-theme 'modus-vivendi))
```

2.12.4 Fonts

```
(use-package fontaine
  :ensure t
  :config
  ;; save file
  (setq fontaine-latest-state-file
    (locate-user-emacs-file "fontaine-latest-state.eld"))
  ;; Set last preset or fall back to desired style from `
  fontaine-presets'.
  (fontaine-set-preset (or (fontaine-restore-latest-preset) 'default
    ))
  ;; The other side of `fontaine-restore-latest-preset'.
  (add-hook 'kill-emacs-hook #'fontaine-store-latest-preset)
  (define-key global-map (kbd "C-c f") #'fontaine-set-preset))
```

```
(use-package rainbow-mode
  :ensure t)
```

2.12.5 Binding for custom select theme function

```
(global-set-key (kbd "M-<f6>") #'bard/select-theme)
```

2.12.6 Mixed pitch faces for writing

```
(use-package mixed-pitch
  :ensure t
  :hook
  (org-mode . mixed-pitch-mode))
```

2.12.7 Spacious padding

```
(use-package spacious-padding
  :ensure t
  :config
  (setq spacious-padding-widths
    '( :internal-border-width 10
      :header-line-width 4
```



```
      :mode-line-width 6
      :tab-width 4
      :right-divider-width 1
      :left-fringe-width 0
      :right-fringe-width 0
      :scroll-bar-width 0))
(spacious-padding-mode t))
```

2.12.8 Provide module

```
(provide 'bard-emacs-theme)
;;; bard-emacs-theme.el ends here
```

2.13 bard-emacs-ui

2.13.1 Whitespace

```
(use-package whitespace
  :ensure nil
  :demand t
  :config
  (setq whitespace-style '(face
                           tabs
                           spaces
                           trailing
                           space-before-tab
                           newline indentation
                           empty space-after-tab
                           space-mark tab-mark))

  :hook
  (prog-mode . whitespace-mode))
```

2.13.2 Line numbers

```
(use-package display-line-numbers
  :ensure nil
  :demand t
  :bind
  (("f12" . display-line-numbers-mode))
  :hook (prog-mode . display-line-numbers-mode)
  :config
  (setq display-line-numbers-type 'relative))
```

2.13.3 Todo keywords

```
(use-package hl-todo
  :ensure t
  :hook
  (prog-mode . hl-todo-mode)
  :config
  (setq hl-todo-highlight-punctuation ":"))
```

2.13.4 Keycasting

```
(use-package keycast
  :ensure t
  :defer t
  :commands (keycast-mode-line-mode keycast-header-line-mode
    keycast-tab-bar-mode keycast-log-mode)
  :init
  (setq keycast-mode-line-format "%2s%k%cR")
  (setq keycast-mode-line-insert-after 'prot-modeline-vc-branch)
  (setq keycast-mode-line-window-predicate '
    mode-line-window-selected-p)
  (setq keycast-mode-line-remove-tail-elements nil)
  :config
  (dolist (input '(self-insert-command org-self-insert-command))
    (add-to-list 'keycast-substitute-alist `(. "Typing"))))
  (dolist (event ("" "<mouse-movement>" "<mouse-2>" "<
    drag-mouse-1>" "<wheel-up>" "<wheel-down>" "<double-wheel-up>" "<
    double-wheel-down>" "<triple-wheel-up>" "<triple-wheel-down>" "<
    wheel-left>" "<wheel-right>" handle-select-window mouse-set-point
    mouse-drag-region))
    (add-to-list 'keycast-substitute-alist `(,event nil nil)))))
```

2.13.5 Provide module

```
(provide 'bard-emacs-ui)
;;; bard-emacs-ui.el ends here
```

2.14 bard-emacs-web

2.14.1 IRC

```
;;; IRC
```

```

(use-package circe
  :ensure t
  :defer t
  :config
  (setq auth-sources '("~/authinfo.gpg"))
  (defun my-fetch-password (&rest params)
    (require 'auth-source)
    (let ((match (car (apply 'auth-source-search params))))
      (if match
          (let ((secret (plist-get match :secret)))
            (if (functionp secret)
                (funcall secret)
                secret))
          (error "Password not found for %S" params))))
  (defun my-nickserv-password (server)
    (my-fetch-password :user "bardman" :machine "irc.libera.chat"))
  (setq circe-network-options
        '(("Libera Chat"
           :nick "bardman"
           ;; :channels ("#emacs" "##anime" "#gentoo")
           :nickserv-password my-nickserv-password))))

```

2.14.2 RSS Feeds

```

(use-package elfeed-protocol
  :ensure t
  :after elfeed
  :config
  (setq elfeed-protocol-enabled-protocols '(fever))
  (elfeed-protocol-enable))
(use-package elfeed
  :ensure t
  :config
  (require 'bard-web)
  (setq elfeed-search-filter "+unread")
  (setq elfeed-feeds '(("fever+https://bard@feeds.bardman.dev/fever/"
                       :api-url "https://feeds.bardman.dev/fever/"
                       :use-authinfo t)))
  :bind
  ("C-c r" . elfeed)

```

```
(:map elfeed-search-mode-map
  ;; C-p for play now
  ("C-c C-p" . bard/play-elfeed-video)
  ;; C-e for EMMS
  ("C-c C-e" . bard/add-video-emms-queue)
  ;; C-w for watch later
  ("C-c C-w" . bard/add-video-watch-later)
  ;; F is for fetch
  ("F" . elfeed-update)))
```

2.14.3 Web Browsing (EWW and external)

```
(use-package eww
  :defer t
  :config
  (setq browse-url-handlers
    '(("wikipedia\\.org" . eww-browse-url)
      ("github" . browse-url-default-browser)
      ("youtube.com" . browse-url-default-browser)
      ("reddit.com" . browse-url-default-browser)))
  ;; shr optimizations
  (setq shr-use-colors nil)
  (setq shr-use-fonts nil)
  (setq shr-max-image-proportion 0.6)
  (setq shr-image-animate nil)
  (setq shr-width fill-column)
  (setq shr-max-width fill-column)
  (setq shr-discard-aria-hidden t)
  (setq shr-cookie-policy nil)
  (defun skip-empty-li (shrtagli dom)
    (when (cddr dom)
      (funcall shrtagli dom)))
  ;; remove empty <li> tags, specifically for wiktionary
  ;; https://nickdrozd.github.io/2022/12/19/wiktionary-advice.html
  (advice-add
    'shr-tag-li :around
    #'skip-empty-li)
  ;; eww
  (setq eww-search-prefix "https://duckduckgo.com/html/?q=")
  (setq eww-history-limit 150)
  (setq eww-use-external-browser-for-content-type
```

```
        "\\`\\(video/\\|audio\\)")
:bind
("C-c w" . eww))
(provide 'bard-emacs-web)
;;; bard-emacs-web.el ends here
```

2.15 bard-emacs-window

2.15.1 Load required libraries

```
(require 'bard-window)
```

```
(use-package emacs
;; configuration for window splits/window sizes
:config
(setq focus-follows-mouse t)
(setq mouse-autoselect-window t)
(setq window-combination-resize t)
(setq even-window-sizes 'height-only)
(setq window-sides-vertical nil)
(setq switch-to-buffer-in-dedicated-window 'pop)
(setq split-height-threshold 80
      split-width-threshold 125)
(setq window-min-height 3
      window-min-width 30))
```

2.15.2 Moving windos

```
(use-package windmove
:bind*
(("C-M-<up>" . windmove-up)
 ("C-M-<right>" . windmove-right)
 ("C-M-<down>" . windmove-down)
 ("C-M-<left>" . windmove-left)
 ("C-M-S-<up>" . windmove-swap-states-up)
 ("C-M-S-<right>" . windmove-swap-states-right)
 ("C-M-S-<down>" . windmove-swap-states-down)
 ("C-M-S-<left>" . windmove-swap-states-left)))
```

2.15.3 Window display rules (`display-buffer-alist`)

Watch [Protesilaos Stavrou's video](#) about this to understand it. My configuration Just Works™, so I don't really get how to change it that much anymore. He can explain it a lot better than I can.

```
(use-package emacs
  :config
  (setq display-buffer-alist
    `((; no window
      ("\\`\\*\\(Async Shell Command\\|typst-ts-compilation\\|
Flycheck error messages\\)\\*\\`"
      (display-buffer-no-window)
      (allow-no-window . t))
      ("\\`\\*\\(Warnings\\|Compile-Log\\|Org Links\\)\\*\\`"
      (display-buffer-no-window)
      (allow-no-window . t))
      ;; bottom side window
      ("\\*Org \\(Select\\|Note\\)\\*"; the `org-capture' key
selection and `org-add-log-note'
      (display-buffer-in-side-window)
      (dedicated . t)
      (side . bottom)
      (slot . 0)
      (window-parameters . ((mode-line-format . none))))
      ;; bottom buffer (NOT side window)
      ((or . ((derived-mode . flymake-diagnostics-buffer-mode)
      (derived-mode . flymake-project-diagnostics-mode)
      (derived-mode . messages-buffer-mode)
      (derived-mode . backtrace-mode)))
      (display-buffer-reuse-mode-window
display-buffer-at-bottom)
      (window-height . 0.3)
      (dedicated . t)
      (preserve-size . (t . t)))
      ;; terminal popups
      (prot-window-shell-or-term-p
      (display-buffer-reuse-mode-window
display-buffer-at-bottom)
      (mode . (shell-mode eshell-mode comint-mode))
      (body-function . prot-window-select-fit-size))
      ("\\magit: .*"
```

```

        (display-buffer-same-window)
        (inhibit-same-window . nil)
        (dedicated . t))
    ("\\*Org Agenda\\*")
    (display-buffer-same-window)
    (inhibit-same-window . nil)
    (dedicated . t))
    ("\\*cfw-calendar\\*")
    (display-buffer-same-window)
    (inhibit-same-window . nil)
    (dedicated . t))
    ("\\*image-dired\\*")
    (display-buffer-reuse-mode-window
display-buffer-in-side-window)
    (side . bottom)
    (window-height . 0.5))
    ("\\*image-dired-display-image\\*")
    (display-buffer-reuse-mode-window
display-buffer-in-side-window)
    (side . right)
    (window-width . 0.35))
    ;; ("\\*Embark Actions\\*")
    ;; (display-buffer-reuse-mode-window
display-buffer-below-selected)
    ;; (window-height . fit-window-to-buffer)
    ;; (window-parameters . ((no-other-window . t)
    ;;                          (mode-line-format . none))))
    ("\\*\\|(Output\\|Register Preview\\|).*"
    (display-buffer-reuse-mode-window
display-buffer-at-bottom))
    ;; below current window
    ("\\(\\*Capture\\*\\|CAPTURE-.*\\|)"
    (display-buffer-in-side-window)
    (dedicated . t)
    (side . bottom)
    (slot . 0)
    (window-parameters . ((mode-line-format . none))))
    ("\\*\\|vc-\\|(incoming\\|outgoing\\|git : \\|).*"
    (display-buffer-reuse-mode-window
display-buffer-below-selected)
    (window-height . 0.1)

```

```

        (dedicated . t)
        (preserve-size . (t . t)))
        ((derived-mode . reb-mode) ; M-x re-builder
         (display-buffer-reuse-mode-window
display-buffer-below-selected)
         (window-height . 4) ; note this is literal lines, not
relative
        (dedicated . t)
        (preserve-size . (t . t)))
        ((or . ((derived-mode . occur-mode)
                 (derived-mode . grep-mode)
                 (derived-mode . Buffer-menu-mode)
                 (derived-mode . log-view-mode)
                 (derived-mode . help-mode) ; See the hooks for `
visual-line-mode'
                 "\\*\\(|Buffer List\\|Occur\\|vc-change-log\\|
eldoc.*\\).*)"
         (prot-window-shell-or-term-p
          ;; ,world-clock-buffer-name
          ))
        (prot-window-display-buffer-below-or-pop)
        (body-function . prot-window-select-fit-size))
        ("\\*\\(|Calendar\\|Bookmark Annotation\\|ert\\).*)"
        (display-buffer-reuse-mode-window
display-buffer-below-selected)
        (dedicated . t)
        (window-height . fit-window-to-buffer))
        ("\\*ispell-top-choices\\*.*)"
        (display-buffer-reuse-mode-window
display-buffer-below-selected)
        (window-height . fit-window-to-buffer))
        )))

```

2.15.4 Frame

1. Keybinds

```

(use-package frame
  :ensure nil
  :bind (("C-x u" . undelete-frame)
         ("C-x f" . other-frame-prefix)

```



```

        ("C-x w t" . tear-off-window)
        ("C-x w c" . clone-indirect-buffer-other-window))
:hook (after-init . undelete-frame-mode))

```

2. Renaming frame

```

(defun bard/rename-frame (frame name)
  "Rename FRAME to NAME."
  (interactive
    (let* ((frames (frame-list))
           (frame (completing-read
                    "Frame: "
                    (mapcar (lambda (f)
                            (frame-parameter f 'name))
                          frames)
                    nil t))
           (frame-obj (seq-find
                      (lambda (f)
                        (equal (frame-parameter f 'name) frame)
                      frames)))
           (list frame-obj
                 (read-string
                  (format "Rename `%s' to: " frame)
                  frame))))
    (modify-frame-parameters frame
      (list (cons 'name name)))))

```

3. Beframe

```

(use-package beframe
  :ensure t
  :config
  (setq beframe-functions-in-frames '(project-prompt-project-dir
                                       notmuch))
  (setq beframe-create-frame-scratch-buffer nil)
  (setq beframe-global-buffers '("*scratch*" "*Messages*" "*
    Backtrace*"))
  (beframe-mode 1)
  (define-key global-map (kbd "C-x f") #'other-frame-prefix)
  (define-key global-map (kbd "C-c b") beframe-prefix-map)
  (define-key global-map (kbd "C-x C-b") #'beframe-buffer-menu)
  (define-key global-map (kbd "C-x B") #'select-frame-by-name)

```

```

(define-key global-map (kbd "C-c b u") #'
  beframe-unassume-current-frame-buffers-selectively)
(define-key global-map (kbd "C-c b a") #'
  beframe-assume-buffers-selectively-all-frames)
;; Consult integration
(defvar consult-buffer-sources)
(declare-function consult--buffer-state "consult")
(with-eval-after-load 'consult
  (defface beframe-buffer
    '((t :inherit font-lock-string-face))
    "Face for `consult' framed buffers.")
  (defun my-beframe-buffer-names-sorted (&optional frame)
    "Return the list of buffers from `beframe-buffer-names'
sorted by visibility.
With optional argument FRAME, return the list of buffers of
FRAME."
    (beframe-buffer-names frame :sort #'
beframe-buffer-sort-visibility))
  (defvar beframe-consult-source
    `( :name      "Frame-specific buffers (current frame)"
      :narrow    ?F
      :category  buffer
      :face      beframe-buffer
      :history   beframe-history
      :items     ,#'my-beframe-buffer-names-sorted
      :action    ,#'switch-to-buffer
      :state     ,#'consult--buffer-state))
    (add-to-list 'consult-buffer-sources 'beframe-consult-source
  )))
(provide 'bard-emacs-window)
;;; bard-emacs-window.el ends here

```

2.15.5 Undoing window actions (**winner-mode**)

```

(use-package winner-mode
  :init
  (winner-mode 1)
  :bind
  (("C-x <right>" . winner-redo)
   ("C-x <left>" . winner-undo)
   ("C-x C-n" . next-buffer))

```

```

("C-x C-p" . previous-buffer)
("C-x <up>" . next-buffer)
("C-x <down>" . previous-buffer)
("C-x w w" . bard/toggle-window-split)))

```

2.15.6 Ibuffer

```

(use-package ibuffer
  :ensure nil
  :config
  (setq ibuffer-default-sorting-mode 'major-mode)
  :hook
  (ibuffer-mode . ibuffer-auto-mode))

```

2.16 bard-emacs-writing

```

(use-package emacs
  :ensure nil
  :demand t
  :bind
  (("C-x i" . insert-char)
   ("M-z" . zap-to-char)
   ("<f10>" . toggle-input-method))
  :config
  ;; Sentence size
  (setq sentence-end-double-space nil)
  ;; Keyboard things
  (setq default-input-method "cyrillic-yawerty")
  (setq default-transient-input-method "cyrillic-yawerty"))
;; spell check
(use-package text-mode
  :ensure nil
  :hook
  (text-mode . flyspell-mode))
;; Tab settings
(use-package emacs
  :config
  (setq tab-always-indent 'complete)
  (setq tab-first-completion 'word-or-paren-or-punct)
  (setq-default tab-width 4
                 indent-tabs-mode nil))

```

2.16.1 Snippets

```
(use-package yasnippet
  :ensure t
  :hook ((LaTeX-mode . yas-minor-mode)
         (post-self-insert . my/yas-try-expanding-auto-snippets))
  :config
  (setq yas-snippet-dirs '("~/emacs.d/snippets"))
  (yas-global-mode t)
  (use-package warnings
    :config
    (cl-pushnew '(yasnippet backquote-change)
                 warning-suppress-types
                 :test 'equal))
  (setq yas-triggers-in-field t)
  ;; Function that tries to autoexpand YaSnippets
  ;; The double quoting is NOT a typo!
  (defun my/yas-try-expanding-auto-snippets ()
    (when (and (boundp 'yas-minor-mode) yas-minor-mode)
      (let ((yas-buffer-local-condition '(require-snippet-condition
                                           . auto)))
        (yas-expand))))))
(use-package yasnippet-capf
  :ensure t
  :after (cape yasnippet)
  :demand t
  :config
  (defun bard/cape-yas-setup ()
    (let* ((capfs (delq t (append completion-at-point-functions '(
      yasnippet-capf)))))
      (super (apply #'cape-capf-super capfs)))
      (setq-local completion-at-point-functions (list super t))))
  (add-hook 'prog-mode-hook #'bard/cape-yas-setup)
  (add-hook 'text-mode-hook #'bard/cape-yas-setup)
  (add-hook 'typst-ts-mode-hook #'bard/cape-yas-setup))
(use-package cdlatex
  :ensure t
  :defer t
  :bind (:map cdlatex-mode-map
              ("" . cdlatex-tab))
  :hook ((LaTeX-mode . turn-on-cdlatex)
         (cdlatex-tab . yas-expand))
```

```

        (cdlatex-tab . cdlatex-in-yas-field))
:config
(use-package yasnippet
  :bind (:map yas-keymap
    (<tab> . yas-next-field-or-cdlatex)
    (TAB . yas-next-field-or-cdlatex))
:config
(defun cdlatex-in-yas-field ()
  ;; Check if we're at the end of the Yas field
  (when-let* ((_ (overlayp yas--active-field-overlay))
    (end (overlay-end yas--active-field-overlay)))
    (if (>= (point) end)
      ;; Call yas-next-field if cdlatex can't expand here
      (let ((s (thing-at-point 'sexp)))
        (unless (and s (assoc (substring-no-properties s)
          cdlatex-command-alist-comb))
          (yas-next-field-or-maybe-expand)
          t))
      ;; otherwise expand and jump to the correct location
      (let (cdlatex-tab-hook minp)
        (setq minp
          (min (save-excursion (cdlatex-tab)
            (point))
            (overlay-end yas--active-field-overlay)))
        (goto-char minp) t))))))
(defun yas-next-field-or-cdlatex nil
  (interactive)
  "Jump to the next Yas field correctly with cdlatex active."
  (if
    (or (bound-and-true-p cdlatex-mode)
      (bound-and-true-p org-cdlatex-mode))
    (cdlatex-tab)
    (yas-next-field-or-maybe-expand))))

```

2.16.2 Denote Note taking

```

(use-package denote
  :ensure t
  :config
  (require 'bard-writing)
  (setq denote-directory "~/Notes/denote/"))

```

```

(setq denote-buffer-name-prefix "[Note] "
  denote-rename-buffer-format "%t %b")
(setq denote-known-keywords
  '("emacs"
    "linux"
    "programming"
    "org"
    "school"
    "language"
    "history"
    "biology"
  ))
(setq denote-templates
  '((blank . "")
    (default . "Related to – ")
    (todo . bard/denote-todo-template)
    (typst-lecture . "#import \"./typst-setups/lecture.typ\":
*\n\n#show: lecture.with(\n  title: \"\", \n  class: \"\", \n
author: \"Daniel Pinkston\", \n)\n\n\")))
(setq denote-save-buffers t)
(setq denote-prompts '(title keywords))
(setq denote-sort-dired-extra-prompts nil)
(setq denote-sort-dired-default-sort-component 'identifier)
(setq denote-sort-dired-default-reverse-sort nil)
;; backlinks sidebar
(setq denote-backlinks-display-buffer-action
  '((display-buffer-in-direction)
    (direction . right)
    (window-width . 0.33)
    (window-height . fit-window-to-buffer)
    (dedicated . t)))
(denote-rename-buffer-mode 1)
(require 'bard-writing)
:hook
(dired-mode . denote-dired-mode)
:bind
(("C-c n n" . denote)
 ("C-c n d" . denote-sort-dired)
 ("C-c n r" . denote-rename-file-using-front-matter)
 ("C-c n k" . denote-rename-file-keywords)
 ("C-c n I" . denote-add-links)

```

```
( "C-c n b" . bard/consult-buffer-notes) ; notes buffer
( "C-c n B" . bard/ibuffer-notes) ; notes buffer but more
(:map dired-mode-map ; rename files in dired with denote
structure
  ("r" . denote-rename-file))))
```

1. Denote org mode extensions

```
(use-package denote-org
  :ensure t
  )
```

2. Denote silo extension

```
(use-package denote-silo
  :ensure t
  :config
  (setq denote-silo-directories '("~/Notes/denote"
                                   "~/Notes/Old Notes/"
                                   "~/Notes/personal/")))
```

3. Denote journal extension

```
(use-package denote-journal
  :ensure t
  :bind
  ("C-c n j" . denote-journal-new-or-existing-entry)
  :config
  (setq denote-journal-directory "~/Notes/denote/journal/")
  (setq denote-journal-title-format 'day-date-month-year-24h))
```

2.16.3 Denote roam

[Denote-roam](#) is my Emacs package that blends org roam and denote into one workflow.

```
(use-package denote-roam
  :ensure nil
  :load-path "~/Code/denote-roam/"
  :bind
  ("C-c n i" . denote-roam-insert-or-create-node) ; node insert
  ("C-c n o" . denote-roam-find-or-create-node) ; node open
  :custom
```

```
(denote-roam-include-journal t)
(denote-roam-directory "~/Notes/denote")
:config
(denote-roam-mode t))
```

2.16.4 Org roam

```
(use-package org-roam
  :ensure t
  :custom
  (org-roam-directory (file-truename "~/Notes/denote"))
  :bind (("C-c n l" . org-roam-buffer-toggle))
  :config
  (cl-defmethod org-roam-node-type ((node org-roam-node))
    "Return the TYPE of NODE based on its filetags."
    (let ((tags (org-roam-node-tags node)))
      (cond
        ((member "bib" tags) "reference")
        ((member "topic" tags) "topic")
        (tags "main")
        (t "root"))))
  ;; (cl-defmethod org-roam-node-denote-id ((node org-roam-node))
  ;;   "Return the denote timestamp ID from NODE's filename, e.g.
  ;;   20251202T154611."
  ;;   (when-let ((file (org-roam-node-file node)))
  ;;     (and (string-match "\\([0-9]\\{8\\}T[0-9]\\{6\\}\\)" file)
  ;;          (match-string 1 file))))
  (cl-defmethod org-roam-node-denote-id ((node org-roam-node))
    "Return reformatted denote timestamp from NODE's filename as
    YYYY-MM-DD HH:MM."
    (when-let ((file (org-roam-node-file node)))
      (and (string-match "\\([0-9]\\{4\\}\\)\\([0-9]\\{2\\}\\)\\([0-9]\\{2\\}\\)T\\([0-9]\\{2\\}\\)\\([0-9]\\{2\\}\\)" file)
           (format "%s-%s-%s %s:%s"
                    (match-string 1 file) ; YYYY
                    (match-string 2 file) ; MM
                    (match-string 3 file) ; DD
                    (match-string 4 file) ; HH
                    (match-string 5 file)))) ; MM
  ;; depends on denote faces :P
  (setq org-roam-node-display-template
```



```

        (concat (propertize "${denote-id:20}" 'face '
denote-faces-date)
                (propertize "${title:60}" 'face 'denote-faces-title
)
                (propertize "${tags:*}" 'face 'denote-faces-keywords
)))
(setq org-roam-db-node-include-function
      (lambda ()
        (not (member "ATTACH" (org-get-tags)))))
(org-roam-db-autosync-mode 1))
(use-package org-roam-ui
  :ensure t
  :bind
  ("C-c n u" . org-roam-ui-open)
  :custom
  (org-roam-ui-open-on-start nil))

```

2.16.5 Org mode annotations in other modes (**org-remark**)

```

(use-package org-remark
  :ensure t
  :config
  (with-eval-after-load 'nov
    (org-remark-nov-mode t))
  (with-eval-after-load 'eww
    (org-remark-eww-mode t))
  :bind
  (:map eww-mode-map
    ("m" . org-remark-mark)
    ("o" . org-remark-open)))

```

2.16.6 Focused writing environment

```

;; Center line scrolling for focused writing
(use-package emacs
  :config
  (define-minor-mode bard/scroll-center-cursor-mode
    "Toggle centered cursor scrolling behavior."
    :init-value nil
    :lighter " S="
    :global nil

```

```

(if bard/scroll-center-cursor-mode
  (setq-local scroll-margin (* (frame-height) 2)
               scroll-conservatively 0
               maximum-scroll-margin 0.5)
  (dolist (local '(scroll-preserve-screen-position
                   scroll-conservatively
                   maximum-scroll-margin
                   scroll-margin))
    (kill-local-variable `,local))))
(defun bard/cursor-centered-p ()
  "Check if `bard/scroll-center-cursor-mode` is currently active."
  (bound-and-true-p bard/scroll-center-cursor-mode))
:bind
(("C-c L" . bard/scroll-center-cursor-mode)))
(use-package olivetti
  :ensure t
  :config
  (setq olivetti-minimum-body-width 90)
  (setq olivetti-recall-visual-line-mode-entry-state t)
  :hook
  ((olivetti-mode-on . (lambda () (olivetti-set-width 90)))
   ))
;; narrowing and focus mode
(use-package logos
  :ensure t
  :config
  (defun logos-reveal-entry ()
    "Reveal Org or Outline entry."
    (cond
      ((and (eq major-mode 'org-mode)
            (org-at-heading-p))
       (org-show-subtree))
      ((or (eq major-mode 'outline-mode)
           (bound-and-true-p outline-minor-mode))
       (outline-show-subtree))))
  (setq logos-outlines-are-pages t)
  (setq logos-outline-regexp-alist
    `((emacs-lisp-mode . "^;;+ ")
      (org-mode . "^\\* +")
      (t . ,(or outline-regexp logos--page-delimiter))))
  (setq-default logos-hide-cursor nil

```

```

        logos-hide-mode-line nil
        logos-hide-header-line t
        logos-hide-buffer-boundaries t
        logos-hide-fringe t
        logos-variable-pitch t
        logos-olivetti t)
(defun bard/logos--recenter-top ()
  "Use `recenter' to reposition the view at the top."
  (unless (derived-mode-p 'prog-mode)
    (recenter 1))) ; Use 0 for the absolute top
:hook
((logos-page-motion . bard/logos--recenter-top))
:hook
((org-mode . logos-focus-mode)
 (markdown-mode . logos-focus-mode))
:bind
(("M-]" . logos-forward-page-dwim)
 ("M-[" . logos-backward-page-dwim)
 ("<f9>" . logos-focus-mode)
 ("C-x n n" . logos-narrow-dwim)))

```

2.16.7 Typst mode and preview

```

(use-package typst-ts-mode
  :ensure t
  :hook (typst-ts-mode . eglot-ensure)
  :config
  (add-to-list 'treesit-language-source-alist
    '(typst "https://github.com/uben0/tree-sitter-typst")
  ))
;; Tinymist for LSP
(with-eval-after-load 'eglot
  (add-to-list 'eglot-server-programs
    '(typst-ts-mode "tinymist")))
(use-package typst-preview
  :ensure t
  :init
  (setq typst-preview-autostart t)
  (setq typst-preview-open-browser-automatically t)
  :config
  (setq typst-preview-browser "zen")

```

```

    typst-preview-invert-colors "never"
    typst-preview-executable "tinymist"
    typst-preview-partial-rendering t)
;; :config
;; (define-key typst-preview-mode-map (kbd "C-c C-j") '
  typst-preview-send-position)
)

```

2.16.8 Citations and bibliography

1. Citar bibliography front end

```

(use-package citar
  :ensure t
  :bind
  ("C-c n c" . citar-open)
  :config
  (setq citar-bibliography '("~/Documents/bib/My Library.bib"))
  (setq org-cite-global-bibliography citar-bibliography
        org-cite-insert-processor 'citar
        org-cite-follow-processor 'citar
        org-cite-activate-processor 'citar)
  (setq citar-notes-paths '("~/Notes/denote"))
  (setq citar-library-paths '("~/Documents/Research Articles/"))
  (defun bard/citar-open-pdf-zathura (file)
    "Open PDF FILE in Zathura."
    (start-process "zathura" nil "zathura" file))
  (setq citar-file-open-functions '(("html" .
    citar-file-open-external)
                                     ("pdf" . bard/
    citar-open-pdf-zathura)
                                     (t . find-file)))
  ;; (setq citar-file-open-functions 'find-file)
  :hook
  (org-mode . citar-capf-setup)
  :bind (("C-c i" . citar-insert-citation)))
(use-package citar-denote
  :ensure t
  :config
  (citar-denote-mode t)
)

```

```
(use-package citar-embark
  :ensure t
  )
```

2. Creating bibliographic entries

```
(defun bard/create-bib-entry ()
  (interactive)
  (let* ((default-file (when (derived-mode-p 'dired-mode)
                           (dired-get-file-for-visit)))
        (file (read-file-name "PDF: "
                              "~/Documents/Research Articles/"
                              default-file nil
                              (when default-file
                                (file-name-nondirectory
                                 default-file))))))
    (find-file "~/Documents/bib/references.bib")
    (goto-char (point-max))
    (yas-expand-snippet
     (format "@article{$1,
author      = {$2},
title       = {$3},
journaltitle = {$4},
year        = {$5},
doi         = {$9},
url         = {$10},
file        = {PDF:%s:application/pdf},
}" file))))
```

3. Citations for typst

```
(use-package citar-typst
  :ensure t
  :hook (typst-ts-mode . citar-typst-mode))
```

2.16.9 Provide module

```
(provide 'bard-emacs-writing)
```

3 Libraries

3.1 bard-calendar

```
(require 'org)
```

3.1.1 Study session program spawning

I use a timer to study sometimes when I really don't lock in. It is a [common lisp script](#) that manages process of a timer called [sowon](#).

```
(defun bard/auto-clock-in ()
  "Automatically clock in when task marked in progress (INPROG),
  and start study session."
  (when (equal (org-get-todo-state) "INPROG")
    (org-clock-in)
    (bard/study-session)))
(defun bard/study-session ()
  "Prompt for study parameters, run study session, and clock out
  when done."
  (interactive)
  (let* ((study-time (read-string "Study time (minutes): "))
        (break-time (read-string "Break time (minutes): "))
        (sessions (read-string "Number of sessions: "))
        (command (format "study %s %s %s" study-time break-time
                          sessions)))
    (process (start-process-shell-command "study-session" "*"
    study*" command)))
  (set-process-sentinel
   process
   (lambda (_proc event)
     (when (string= event "finished\n")
       (progn (org-clock-out)
              (pop-to-buffer-same-window "todo.org"))))))))
```

3.1.2 Org clock

```
(defun bard/org-clock-report ()
  "Generate an org clock report in a separate org buffer."
  (interactive)
  (bard/new-org-buffer)
  (org-clock-report))
(defun bard/org-clock-update-mode-line ()
  "Update the modeline, not sure why I have this."
  (interactive))
```

```

    (setq org-mode-line-string nil)
    (force-mode-line-update))
(defun bard/org-clock-task-string ()
  "Return a simplified org clock task string.
Used in FVWM3 configuration to show clocked task in FVWMbuttons."
  (if (and (boundp 'org-mode-line-string)
           (not (string-equal "" org-mode-line-string))
           org-mode-line-string)
      (substring-no-properties org-mode-line-string)
      "No task clocked in"))

```

3.1.3 Open the calendar in its own window

I use this in window manager status bar to open a popup window to view the dates on the calendar.

```

(defun bard/open-calendar ()
  "Opens calendar as only window"
  (interactive)
  (calendar)
  (delete-other-windows))

```

3.1.4 Org agenda

```

(defun bard/choose-agenda ()
  "For viewing my custom agenda"
  (interactive)
  (let ((agenda-views '("Default" "Monthly" "Yearly")))
    (setq chosen-view (completing-read "Choose an agenda view: "
                                       agenda-views))
    (cond
     ((string= chosen-view "Yearly")
      (org-agenda nil "Y"))
     ((string= chosen-view "Monthly")
      (org-agenda nil "M"))
     ((string= chosen-view "Default")
      (org-agenda nil "D")))))
(defun bard/default-agenda ()
  "For viewing my custom agenda"
  (interactive)
  (org-agenda nil "D"))

```

3.1.5 Provide library

```
(provide 'bard-calendar)
;;; bard-calendar.el ends here
```

3.2 bard-compile

Code taken from [here](#). I need this to solve a problem I had with pros-cli back when I did robotics in high school. The program had a bunch of color codes that weren't properly displayed in the compilation output. I really like using `M-x compile` for C/C++ development specifically.

```
(require 'ansi-color)
(defun endless/colorize-compilation ()
  "Colorize from `compilation-filter-start' to `point'."
  (let ((inhibit-read-only t))
    (ansi-color-apply-on-region
     compilation-filter-start (point))))
(add-hook 'compilation-filter-hook
          #'endless/colorize-compilation)
;; Stolen from (https://oleksandrmanzyuk.wordpress.com/2011/11/05/
  better-emacs-shell-part-i/)
(defun regexp-alternatives (regexps)
  "Return the alternation of a list of regexps."
  (mapconcat (lambda (regexp)
                (concat "\\(?:" regexp "\\)"))
              regexps "\\|"))
(defvar non-sgr-control-sequence-regexp nil
  "Regex that matches non-SGR control sequences.")
(setq non-sgr-control-sequence-regexp
      (regexp-alternatives
       '(;; icon name escape sequences
         "\\033\\[[0-2];.*?\\007"
         ;; non-SGR CSI escape sequences
         "\\033\\[[\\0-9;]*[^0-9;m]"
         ;; noop
         "\\012\\033\\[[2K\\033\\[[1F"
       )))
(defun filter-non-sgr-control-sequences-in-region (begin end)
  (save-excursion
    (goto-char begin)
    (while (re-search-forward
            non-sgr-control-sequence-regexp end t)
```



```

        (replace-match ""))))
(defun filter-non-sgr-control-sequences-in-output (ignored)
  (let ((start-marker
        (or comint-last-output-start
            (point-min-marker))))
    (end-marker
      (process-mark
        (get-buffer-process (current-buffer)))))
    (filter-non-sgr-control-sequences-in-region
      start-marker
      end-marker)))
(add-hook 'comint-output-filter-functions
  'filter-non-sgr-control-sequences-in-output)
(provide 'bard-compile)

```

3.3 bard-email

Word for word copy of Protesilaos notmuch library. Email in Emacs is already complicated enough, and I had enough of a hard time without his help.

```

(require 'prot-common)
(eval-when-compile (require 'cl-lib))
(defgroup prot-notmuch ()
  "Extensions for notmuch.el."
  :group 'notmuch)
(defcustom prot-notmuch-delete-tag "del"
  "Single tag that applies to mail marked for deletion.
This is used by `prot-notmuch-delete-mail'."
  :type 'string
  :group 'prot-notmuch)
(defcustom prot-notmuch-mark-delete-tags
  `((, (format "+%s" prot-notmuch-delete-tag) "-inbox" "-unread"))
  "List of tags to mark for deletion.
To actually delete email, refer to `prot-notmuch-delete-mail'."
  :type '(repeat string)
  :group 'prot-notmuch)
(defcustom prot-notmuch-mark-flag-tags '("+flag" "-unread")
  "List of tags to mark as important (flagged).
This gets the `notmuch-tag-flagged' face, if that is specified in
`notmuch-tag-formats'."

```

```

:type '(repeat string)
:group 'prot-notmuch)
(defcustom prot-notmuch-mark-spam-tags '("+spam" "-inbox" "-unread")
  "List of tags to mark as spam."
:type '(repeat string)
:group 'prot-notmuch)
(autoload 'notmuch-interactive-region "notmuch")
(autoload 'notmuch-tag-change-list "notmuch")
(autoload 'notmuch-search-next-thread "notmuch")
(autoload 'notmuch-search-tag "notmuch")
(defmacro prot-notmuch-search-tag-thread (name tags)
  "Produce NAME function parsing TAGS."
  (declare (indent defun))
  `(defun ,name (&optional untag beg end)
    , (format
      "Mark with `%s' the currently selected thread.
Operate on each message in the currently selected thread.  With
optional BEG and END as points delimiting a region that
encompasses multiple threads, operate on all those messages
instead.
With optional prefix argument (\\[universal-argument]) as UNTAG,
reverse the application of the tags.
This function advances to the next thread when finished."
      tags)
    (interactive (cons current-prefix-arg (
notmuch-interactive-region)))
    (when ,tags
      (notmuch-search-tag
        (notmuch-tag-change-list ,tags untag) beg end))
    (when (eq beg end)
      (notmuch-search-next-thread))))
(prot-notmuch-search-tag-thread
  prot-notmuch-search-delete-thread
  prot-notmuch-mark-delete-tags)
(prot-notmuch-search-tag-thread
  prot-notmuch-search-flag-thread
  prot-notmuch-mark-flag-tags)
(prot-notmuch-search-tag-thread
  prot-notmuch-search-spam-thread
  prot-notmuch-mark-spam-tags)
(defmacro prot-notmuch-show-tag-message (name tags)

```

```

"Produce NAME function parsing TAGS."
(declare (indent defun))
`(defun ,name (&optional untag)
  ,(format
    "Apply `%s' to message.
With optional prefix argument (\\[universal-argument]) as UNTAG,
reverse the application of the tags."
    tags)
  (interactive "P")
  (when ,tags
    (apply 'notmuch-show-tag-message
            (notmuch-tag-change-list ,tags untag)))))
(prot-notmuch-show-tag-message
 prot-notmuch-show-delete-message
 prot-notmuch-mark-delete-tags)
(prot-notmuch-show-tag-message
 prot-notmuch-show-flag-message
 prot-notmuch-mark-flag-tags)
(prot-notmuch-show-tag-message
 prot-notmuch-show-spam-message
 prot-notmuch-mark-spam-tags)
(defun prot-notmuch-delete-mail ()
  "Permanently delete mail marked as `prot-notmuch-delete-mail'.
Prompt for confirmation before carrying out the operation.
Do not attempt to refresh the index. This will be done upon the
next invocation of 'notmuch new'."
  (interactive)
  (let* ((del-tag prot-notmuch-delete-tag)
        (count
         (string-to-number
          (with-temp-buffer
            (shell-command
             (format "notmuch count tag:%s" prot-notmuch-delete-tag
                    ) t)
            (buffer-substring-no-properties (point-min) (1- (
point-max)))))))
        (mail (if (> count 1) "mails" "mail"))))
    (unless (> count 0)
      (user-error "No mail marked as `%s'" del-tag))
    (when (yes-or-no-p
           (format "Delete %d %s marked as `%s'?" count mail del-tag

```

```

    ))
    (shell-command
      (format "notmuch search --output=files --format=text0 tag:%s
| xargs -r0 rm" del-tag)
      t))))
(provide 'bard-email)

```

3.4 bard-embark

Main embark configuration found in `bard-emacs-modules/bard-emacs-embark.el`. This is another one of the files I borrowed from Protesilaos, but renamed/modified so I can remember.

```

(require 'embark)
(defvar-keymap bard-embark-general-map
  :parent embark-general-map
  "i" #'embark-insert
  "w" #'embark-copy-as-kill
  "E" #'embark-export
  "S" #'embark-collect
  "A" #'embark-act-all
  "DEL" #'delete-region)
(defvar-keymap bard-embark-url-map
  :parent embark-general-map
  "b" #'browse-url
  "d" #'embark-download-url
  "e" #'eww)
(defvar-keymap bard-embark-buffer-map
  :parent embark-general-map
  "k" #'bard-simple-kill-buffer
  "o" #'switch-to-buffer-other-window
  "e" #'ediff-buffers)
(add-to-list 'embark-post-action-hooks (list '
  bard-simple-kill-buffer 'embark--restart))
(defvar-keymap bard-embark-file-map
  :parent embark-general-map
  "f" #'find-file
  "j" #'embark-dired-jump
  "c" #'copy-file
  "e" #'ediff-files)

```

```

(defvar-keymap bard-embark-identifier-map
  :parent embark-general-map
  "h" #'display-local-help
  "." #'xref-find-definitions
  "o" #'occur)
(defvar-keymap bard-embark-command-map
  :parent embark-general-map
  "h" #'describe-command
  "." #'embark-find-definition)
(defvar-keymap bard-embark-expression-map
  :parent embark-general-map
  "e" #'pp-eval-expression
  "m" #'pp-macroexpand-expression)
(defvar-keymap bard-embark-function-map
  :parent embark-general-map
  "h" #'describe-function
  "." #'embark-find-definition)
(defvar-keymap bard-embark-package-map
  :parent embark-general-map
  "h" #'describe-package
  "i" #'package-install
  "d" #'package-delete
  "r" #'package-reinstall
  "b" #'embark-browse-package-url
  "w" #'embark-save-package-url)
(defvar-keymap bard-embark-symbol-map
  :parent embark-general-map
  "h" #'describe-symbol
  "." #'embark-find-definition)
(defvar-keymap bard-embark-variable-map
  :parent embark-general-map
  "h" #'describe-variable
  "." #'embark-find-definition)
(defvar-keymap bard-embark-region-map
  :parent embark-general-map
  "a" #'align-regexp
  "D" #'delete-duplicate-lines
  "f" #'flush-lines
  "i" #'epa-import-keys-region
  "d" #'epa-decrypt-armor-in-region
  "r" #'repunctuate-sentences

```

```

"s" #'sort-lines
"u" #'untabify)
;; The minimal indicator shows cycling options, but I have no use
;; for those. I want it to be silent.
(defun bard-embark-no-minimal-indicator ())
(advice-add #'embark-minimal-indicator :override #'
  bard-embark-no-minimal-indicator)
(provide 'bard-embark)

```

3.5 bard-eshell

```

(require 'cl-lib)
(require 'eshell)

```

3.5.1 Eshell aliases

```

;; aliases
(setq bard/eshell-aliases
  '((g . magit)
    (gl . magit-log)
    (d . dired)
    (o . find-file)
    (oo . find-file-other-window)
    (l . (lambda () (eshell/ls '-la)))
    (eshell/clear . eshell/clear-scrollback)))
(mapc (lambda (alias)
  (defalias (car alias) (cdr alias)))
  bard/eshell-aliases)

```

3.5.2 Prot-eshell code

I believe Protesilaos has stopped using eshell, but I went back in his git history to the time when he used it in a video and took some custom elisp from there.

```

(defun prot-eshell--cd (dir)
  "Routine to cd into DIR."
  (delete-region eshell-last-output-end (point-max))
  (when (> eshell-last-output-end (point))
    (goto-char eshell-last-output-end))
  (insert-and-inherit "cd " (eshell-quote-argument dir))
  (eshell-send-input))
(defun prot-eshell-complete-recent-dir (dir &optional arg)

```

```

    "Switch to a recent Eshell directory.
When called interactively, DIR is selected with completion from
the elements of `eshell-last-dir-ring'.
With optional ARG prefix argument (\\[universal-argument]) also
open the directory in a `dired' buffer."
    (interactive
      (list
        (if-let ((dirs (ring-elements eshell-last-dir-ring)))
          (completing-read "Switch to recent dir: " dirs nil t)
          (user-error "There is no Eshell history for recent directories
            ")))
        current-prefix-arg))
    (prot-eshell--cd dir)
;; UPDATE 2022-01-04 10:48 +0200: The idea for `dired-other-window
'
;; was taken from Sean Whitton's `spw/eshell-cd-recent-dir'.
Check
;; Sean's dotfiles: <https://git.spwhitton.name/dotfiles>.
    (when arg
      (dired-other-window dir)))
(defun bard/eshell-find-file-at-point ()
  "Run `find-file` to find file"
  (interactive)
  (let ((file (ffap-file-at-point)))
    (if file
      (find-file file)
      (user-error "No file at point"))))
(defcustom prot-eshell-output-buffer "*Exported Eshell output*"
  "Name of buffer with the last output of Eshell command.
Used by `prot-eshell-export'."
  :type 'string
  :group 'prot-eshell)
(defcustom prot-eshell-output-delimiter "* * *"
  "Delimiter for successive `prot-eshell-export' outputs.
This is formatted internally to have newline characters before
and after it."
  :type 'string
  :group 'prot-eshell)
(defun prot-eshell--command-prompt-output ()
  "Capture last command prompt and its output."
  (let ((beg (save-excursion

```

```

                (goto-char (eshell-beginning-of-input))
                (goto-char (point-at-bol))))))
    (when (derived-mode-p 'eshell-mode)
      (buffer-substring-no-properties beg (eshell-end-of-output))))))
;;;###autoload
(defun prot-eshell-export ()
  "Produce a buffer with output of the last Eshell command.
If `prot-eshell-output-buffer' does not exist, create it. Else
append to it, while separating multiple outputs with
`prot-eshell-output-delimiter'."
  (interactive)
  (let ((eshell-output (prot-eshell--command-prompt-output)))
    (with-current-buffer (get-buffer-create
      prot-eshell-output-buffer)
      (let ((inhibit-read-only t))
        (goto-char (point-max))
        (unless (eq (point-min) (point-max))
          (insert (format "\n%s\n\n" prot-eshell-output-delimiter)))
        (goto-char (point-at-bol))
        (insert eshell-output)
        (switch-to-buffer-other-window (current-buffer))))))
  (defgroup bard-eshell-faces nil
    "Faces for my custom modeline."
    :group 'prot-eshell-faces)
  (defface bard-eshell-highlight-yellow-bg
    '(((default :inherit (bold prot-modeline-indicator-button))
      (((class color) (min-colors 88) (background light))
       :background "#805000" :foreground "white")
      (((class color) (min-colors 88) (background dark))
       :background "#ffc800" :foreground "black")
      (t :background "yellow" :foreground "black")))
    "Face for modeline indicators with a background."
    :group 'bard-eshell-faces)
  (defun prot-eshell-narrow-output-highlight-regexp (regexp)
    "Narrow to last command output and highlight REGEXP."
    (interactive
      (list (read-regexp "Regex to highlight" nil '
        prot-eshell--output-highlight-history)))
    (narrow-to-region (eshell-beginning-of-output)
      (eshell-end-of-output))
    (goto-char (point-min)))

```



```

(highlight-regexp regexp 'prot-eshell-highlight-yellow-bg)
(message "%s to last output and highlighted '%s'"
      (propertyize "Narrowed" 'face 'bold)
      (propertyize regexp 'face 'italic)))
(defun select-or-create (arg)
  "Commentary ARG."
  (if (string= arg "New eshell")
      (eshell t)
      (switch-to-buffer arg)))
(defun eshell-switcher (&optional arg)
  "Commentary ARG."
  (interactive)
  (let* (
    (buffers (cl-remove-if-not (lambda (n) (eq (
buffer-local-value 'major-mode n) 'eshell-mode)) (buffer-list)) )
    (names (mapcar (lambda (n) (buffer-name n)) buffers))
    (num-buffers (length buffers) )
    (in-eshellp (eq major-mode 'eshell-mode)))
    (cond ((eq num-buffers 0) (eshell (or arg t)))
          ((not in-eshellp) (switch-to-buffer (car buffers)))
          (t (select-or-create (completing-read "Select Shell:" (
cons "New eshell" names)))))))

```

3.5.3 Prompt configuration

```

;; taken from https://github.com/karthink/.emacs.d/blob/master/lisp/
  setup-shells.el
(use-package eshell
  :defer
  :config
  (setq eshell-prompt-regexp "^.* λ "
        eshell-prompt-function #'bard/eshell-default-prompt-fn)
  (defun bard/eshell-default-prompt-fn ()
    "Generate the prompt string for eshell. Use for `
    eshell-prompt-function'."
    (concat (if (bobp) "" "\n")
      (let ((pwd (eshell/pwd)))
        (propertyize (if (equal pwd "~")
                          pwd
                          (abbreviate-file-name pwd))
                      'face 'bard/eshell-prompt-pwd))

```

```

        (propertize (bard/eshell--current-git-branch)
                    'face 'bard/eshell-prompt-git-branch)
        (propertize " λ" 'face (if (zerop
eshell-last-command-status) 'success 'error))
        " "))
(defsubst bard/eshell--current-git-branch ()
  ;; TODO Refactor me
  (cl-destructuring-bind (status . output)
    (with-temp-buffer (cons
                      (or (call-process "git" nil t nil "
symbolic-ref" "-q" "--short" "HEAD")
                        (call-process "git" nil t nil "
describe" "--all" "--always" "HEAD")
                      -1)
                      (string-trim (buffer-string)))))
  (if (equal status 0)
      (format " [%s]" output)
      ""))
(defface bard/eshell-prompt-pwd '((t (:inherit
font-lock-keyword-face)))
  "TODO"
  :group 'eshell)
(defface bard/eshell-prompt-git-branch '((t (:inherit
font-lock-builtin-face)))
  "TODO"
  :group 'eshell))

```

3.5.4 Provide library

```
(provide 'bard-eshell)
```

3.6 bard-media

3.6.1 Load required libraries

```

(require 'cl-lib)
(require 'seq)
(require 'emms)
(require 'image-dired)
(require 'dired-x)

```

3.6.2 EMMS Convenience functions

Somewhat related to my Elfeed/EMMS Youtube Code.

```
(defun bard/play-youtube-video ()
  "Play the YouTube URL at point or prompt for one if none is found.
  "
  (interactive)
  (let* ((url-at-point (thing-at-point 'url t))
         (url (if (and url-at-point
                        (string-match-p "https?://\\(www\\.\\.\\.\\)?\\(youtube\\.\\.com\\|youtu\\.\\.be\\.\\.\\.\\)" url-at-point))
                   url-at-point
                   (read-string "Enter YouTube URL: "))))
    (if (and url (string-match-p "https?://\\(www\\.\\.\\.\\)?\\(youtube\\.\\.com\\|youtu\\.\\.be\\.\\.\\.\\)" url))
        (async-shell-command (format "mpv '%s'" url))
        (message "The URL is not a valid YouTube link: %s" url)))
  )

(defun bard/save-emms-watch-later ()
  "Save the current EMMS playlist to `bard/watch-later-file` using `bard/emms-playlist-format`."
  (interactive)
  (when (and bard/watch-later-file bard/emms-playlist-format)
    (emms-playlist-save bard/emms-playlist-format bard/watch-later-file)
    (message "Playlist saved to %s" bard/watch-later-file)))
  )

(defun bard/emms-download-current-video (destination)
  "Download the currently playing EMMS video and move it to DESTINATION."
  (interactive "DSelect destination directory: ")
  (require 'emms)
  (let* ((track (emms-playlist-current-selected-track))
         (url (emms-track-get track 'name))
         (default-directory (file-name-as-directory
                             temporary-file-directory))
         (downloader (executable-find "yt-dlp"))
         (output-template "%(title)s.%(ext)s"))
    (unless downloader
      (error "yt-dlp or youtube-dl is not installed or not in PATH"))
    )
  (unless (string-match-p "^https?://" url)
    (error "Current track is not a valid video URL"))
  )
```

```

(message "Downloading video from: %s" url)
(make-process
 :name "emms-yt-dlp"
 :buffer "*emms-yt-dlp*"
 :command (list downloader "-o" output-template url)
 :sentinel (lambda (proc event)
              (cond
               ((string= event "finished\n")
                (let* ((downloaded-file (car (directory-files
default-directory t ".*\\(mp4\\|mkv\\|webm\\)\\$" 'time)))
                      (target-path (expand-file-name (
file-name-nondirectory downloaded-file) destination)))
                  (rename-file downloaded-file target-path t)
                  (message "Video saved to: %s" target-path)))
               ((string-match-p "exited abnormally" event)
                (message "Download failed! See *emms-yt-dlp* for
details"))))))))

```

3.6.3 Interacting with nsxiv image browser through elisp

I made a video about this workflow [here](#).

```

(defun bard/image-browser-choose (directory)
  "Open nsxiv in thumbnail mode on DIRECTORY.
Asks the user whether to enable recursive mode and whether to output
marked files to a buffer."
  (interactive "DSelect directory: ")
  (let* ((recursive (if (y-or-n-p "Recursive searching? ") "-r" ""))
         (stdout (if (y-or-n-p "Output marked files to buffer? ") "-o" ""))
         (full-dir (expand-file-name directory))
         (args (remove "" (list "nsxiv" "-t" stdout recursive
full-dir))))
    ;; Pre-clear the output buffer if needed
    (when (string= stdout "-o")
      (with-current-buffer (get-buffer-create "*nsxiv*")
        (read-only-mode 0)
        (erase-buffer)))
    (message "Running: %s" (string-join args " "))
    (let ((process (apply #'start-process "nsxiv" "*nsxiv*" args)))
      (when (string= stdout "-o")

```

```

(set-process-sentinel
 process
 (lambda (proc event)
  (when (string= event "finished\n")
   (with-current-buffer "*nsxiv*"
    (read-only-mode nil)
    (goto-char (point-min)))
   ;; Read marked files
   (let ((files (with-current-buffer "*nsxiv*"
                   (split-string (buffer-string) "\n" t))))
    (bard/open-marked-in-dired files))))
  (pop-to-buffer "*nsxiv*")))))
(defun bard/open-marked-in-dired (files)
  "Open a list of FILES in an interactive Dired buffer."
  (if (and files (listp files))
      (dired (cons "*nsxiv-marked*" files))
      (message "No valid files to show in Dired.")))
(defun bard/image-browser-marked ()
  "Open nsxiv on the marked files in Dired."
  (let* ((files (dired-get-marked-files))
         (stdout (if (y-or-n-p "Output marked files to buffer? ") "-o" "")))
    (message "Opening marked files: %s" (string-join files ", "))
    (when (string= stdout "-o")
      (with-current-buffer (get-buffer-create "*nsxiv*")
        (read-only-mode 0)
        (erase-buffer)))
    ;; Corrected apply call:
    (let ((process (apply #'start-process
                          "nsxiv"
                          "*nsxiv*"
                          (append (list "nsxiv" "-t" stdout) files))
      ))
      (when (string= stdout "-o")
        (set-process-sentinel
         process
         (lambda (proc event)
          (when (string= event "finished\n")
            (with-current-buffer "*nsxiv*"
              (read-only-mode nil)
              (goto-char (point-min))))

```

```

                (let ((files (with-current-buffer "*nsxiv*"
                               (split-string (buffer-string) "\n" t))))
                    (bard/open-marked-in-dired files))))
                (pop-to-buffer "*nsxiv*")))))
(defun bard/image-browser ()
  "Open nsxiv in a context-sensitive way:
- If in Dired with marked files, open those with nsxiv.
- If in Dired with no marked files, prompt for a directory.
- If not in Dired, prompt for a directory."
  (interactive)
  (cond
   ;; In Dired and files are marked
   ((and (derived-mode-p 'dired-mode)
        (< 1 (length (dired-get-marked-files)))))
    (message "Opening marked files from Dired...")
    (bard/image-browser-marked))
   ;; In Dired but no marked files
   ((derived-mode-p 'dired-mode)
    (message "No files marked in Dired. Prompting for directory...")
    (call-interactively #'bard/image-browser-choose))
   ;; Not in Dired
   (t
    (message "Not in Dired. Prompting for directory...")
    (call-interactively #'bard/image-browser-choose)))
  (provide 'bard-media.el))

```

3.7 bard-modeline

This is another one of prot's libraries that I copied and modified a long time ago. The modifications this time were quite insignificant, but I did add some custom modules that I wanted.

```

(require 'prot-common)
(defgroup prot-modeline nil
  "Custom modeline that is stylistically close to the default."
  :group 'mode-line)
(defgroup prot-modeline-faces nil
  "Faces for my custom modeline."
  :group 'prot-modeline)
(defcustom prot-modeline-string-truncate-length 9
  "String length after which truncation should be done in small
  windows.")

```

```

:type 'natnum)
(defun mode-line-window-selected-p ()
  "Return non-nil if we're updating the mode line for the selected
  window.
  This function is meant to be called in `:eval' mode line
  constructs to allow altering the look of the mode line depending
  on whether the mode line belongs to the currently selected window
  or not."
  (let ((window (selected-window)))
    (or (eq window (old-selected-window))
        (and (minibuffer-window-active-p (minibuffer-window))
              (with-selected-window (minibuffer-window)
                (eq window (minibuffer-selected-window)))))))
;;; Faces
(defface prot-modeline-indicator-button nil
  "Generic face used for indicators that have a background.
  Modify this face to, for example, add a :box attribute to all
  relevant indicators (combines nicely with my `spacious-padding'
  package).")
(defface prot-modeline-indicator-red
  '((default :inherit bold)
    (((class color) (min-colors 88) (background light))
     :foreground "#880000")
    (((class color) (min-colors 88) (background dark))
     :foreground "#ff9f9f")
    (t :foreground "red"))
  "Face for modeline indicators (e.g. see my `notmuch-indicator')."
  :group 'prot-modeline-faces)
(defface prot-modeline-indicator-red-bg
  '((default :inherit (bold prot-modeline-indicator-button))
    (((class color) (min-colors 88) (background light))
     :background "#aa1111" :foreground "white")
    (((class color) (min-colors 88) (background dark))
     :background "#ff9090" :foreground "black")
    (t :background "red" :foreground "black"))
  "Face for modeline indicators with a background."
  :group 'prot-modeline-faces)
(defface prot-modeline-indicator-green
  '((default :inherit bold)
    (((class color) (min-colors 88) (background light))
     :foreground "#005f00")

```

```

      (((class color) (min-colors 88) (background dark))
       :foreground "#73fa7f")
      (t :foreground "green"))
"Face for modeline indicators (e.g. see my `notmuch-indicator')."
:group 'prot-modeline-faces)
(defface prot-modeline-indicator-green-bg
 '((default :inherit (bold prot-modeline-indicator-button))
  (((class color) (min-colors 88) (background light))
   :background "#207b20" :foreground "white")
  (((class color) (min-colors 88) (background dark))
   :background "#77d077" :foreground "black")
  (t :background "green" :foreground "black")))
"Face for modeline indicators with a background."
:group 'prot-modeline-faces)
(defface prot-modeline-indicator-yellow
 '((default :inherit bold)
  (((class color) (min-colors 88) (background light))
   :foreground "#6f4000")
  (((class color) (min-colors 88) (background dark))
   :foreground "#f0c526")
  (t :foreground "yellow")))
"Face for modeline indicators (e.g. see my `notmuch-indicator')."
:group 'prot-modeline-faces)
(defface prot-modeline-indicator-yellow-bg
 '((default :inherit (bold prot-modeline-indicator-button))
  (((class color) (min-colors 88) (background light))
   :background "#805000" :foreground "white")
  (((class color) (min-colors 88) (background dark))
   :background "#ffc800" :foreground "black")
  (t :background "yellow" :foreground "black")))
"Face for modeline indicators with a background."
:group 'prot-modeline-faces)
(defface prot-modeline-indicator-blue
 '((default :inherit bold)
  (((class color) (min-colors 88) (background light))
   :foreground "#00228a")
  (((class color) (min-colors 88) (background dark))
   :foreground "#88bfff")
  (t :foreground "blue")))
"Face for modeline indicators (e.g. see my `notmuch-indicator')."
:group 'prot-modeline-faces)

```



```

(defface prot-modeline-indicator-blue-bg
  '((default :inherit (bold prot-modeline-indicator-button))
    (((class color) (min-colors 88) (background light))
      :background "#0000aa" :foreground "white")
    (((class color) (min-colors 88) (background dark))
      :background "#77aaff" :foreground "black")
    (t :background "blue" :foreground "black")))
"Face for modeline indicators with a background."
:group 'prot-modeline-faces)
(defface prot-modeline-indicator-magenta
  '((default :inherit bold)
    (((class color) (min-colors 88) (background light))
      :foreground "#6a1aaf")
    (((class color) (min-colors 88) (background dark))
      :foreground "#e0a0ff")
    (t :foreground "magenta")))
"Face for modeline indicators (e.g. see my `notmuch-indicator')."
:group 'prot-modeline-faces)
(defface prot-modeline-indicator-magenta-bg
  '((default :inherit (bold prot-modeline-indicator-button))
    (((class color) (min-colors 88) (background light))
      :background "#6f0f9f" :foreground "white")
    (((class color) (min-colors 88) (background dark))
      :background "#e3a2ff" :foreground "black")
    (t :background "magenta" :foreground "black")))
"Face for modeline indicators with a background."
:group 'prot-modeline-faces)
(defface prot-modeline-indicator-cyan
  '((default :inherit bold)
    (((class color) (min-colors 88) (background light))
      :foreground "#004060")
    (((class color) (min-colors 88) (background dark))
      :foreground "#30b7cc")
    (t :foreground "cyan")))
"Face for modeline indicators (e.g. see my `notmuch-indicator')."
:group 'prot-modeline-faces)
(defface prot-modeline-indicator-cyan-bg
  '((default :inherit (bold prot-modeline-indicator-button))
    (((class color) (min-colors 88) (background light))
      :background "#006080" :foreground "white")
    (((class color) (min-colors 88) (background dark))

```

```

      :background "#40c0e0" :foreground "black")
    (t :background "cyan" :foreground "black"))
  "Face for modeline indicators with a background."
  :group 'prot-modeline-faces)
;;; Common helper functions
(defun prot-modeline--string-truncate-p (str)
  "Return non-nil if STR should be truncated."
  (and (prot-common-window-small-p)
        (> (length str) prot-modeline-string-truncate-length)
        (not (one-window-p :no-minibuffer))))
(defun prot-modeline--truncate-p ()
  "Return non-nil if truncation should happen.
This is a more general and less stringent variant of
`prot-modeline--string-truncate-p'."
  (and (prot-common-window-small-p)
        (not (one-window-p :no-minibuffer))))
(defun prot-modeline-string-truncate (str)
  "Return truncated STR, if appropriate, else return STR.
Truncation is done up to `prot-modeline-string-truncate-length'."
  (if (prot-modeline--string-truncate-p str)
      (concat (substring str 0 prot-modeline-string-truncate-length)
              "...")
      str))
(defun prot-modeline-string-truncate-end (str)
  "Like `prot-modeline-string-truncate' but truncate from STR
beginning."
  (if (prot-modeline--string-truncate-p str)
      (concat "... " (substring str (-
        prot-modeline-string-truncate-length)))
      str))
(defun prot-modeline--first-char (str)
  "Return first character from STR."
  (substring str 0 1))
(defun prot-modeline-string-abbreviate (str)
  "Abbreviate STR individual hyphen or underscore separated words.
Also see `prot-modeline-string-abbreviate-but-last'."
  (if (prot-modeline--string-truncate-p str)
      (mapconcat #'prot-modeline--first-char (split-string str "[_-]"
        t) "-")
      str))
(defun prot-modeline-string-abbreviate-but-last (str nthlast)

```

```

"Abbreviate STR, keeping NTHLAST words intact.
Also see `prot-modeline-string-abbreviate'."
(if (prot-modeline--string-truncate-p str)
    (let* ((all-strings (split-string str "[-]"))
           (nbutlast-strings (nbutlast (copy-sequence all-strings)
                                         nthlast))
           (last-strings (nreverse (ntake nthlast (nreverse (
copy-sequence all-strings))))))
          (first-component (mapconcat #'prot-modeline--first-char
                                         nbutlast-strings "-"))
          (last-component (mapconcat #'identity last-strings "-"))
          ))
      (if (string-empty-p first-component)
          last-component
          (concat first-component "-" last-component)))
    str))
;;; Keyboard macro indicator
(defvar-local prot-modeline-kbd-macro
  '(:eval
    (when (and (mode-line-window-selected-p) defining-kbd-macro)
      (propertize " KMacro " 'face '
prot-modeline-indicator-blue-bg)))
  "Mode line construct displaying `mode-line-defining-kbd-macro'.
Specific to the current window's mode line.")
;;; Narrow indicator
(defvar-local prot-modeline-narrow
  '(:eval
    (when (and (mode-line-window-selected-p)
                (buffer-narrowed-p)
                (not (derived-mode-p 'Info-mode 'help-mode '
special-mode 'message-mode)))
      (propertize " Narrow " 'face '
prot-modeline-indicator-cyan-bg)))
  "Mode line construct to report the multilingual environment.")
;;; Centered cursor indicator
(defvar-local bard-modeline-centered-cursor
  '(:eval
    (when (and (mode-line-window-selected-p)
                (bard/cursor-centered-p)
                (not (derived-mode-p 'Info-mode 'help-mode '
special-mode 'message-mode)))

```

```

        (propertize " Center " 'face '
        prot-modeline-indicator-yellow-bg)))
    "Mode line construct to report the multilingual environment.")
;; FIXME: Combine these two functions one day...
(defvar-local bard-modeline-ryo-modal-normal
  '(:eval
    (when (and (mode-line-window-selected-p)
                (not (bard/ryo-insert-p))
                (not (derived-mode-p 'Info-mode 'help-mode '
special-mode 'message-mode))))
      (propertize "<N>" 'face 'prot-modeline-indicator-magenta-bg)
    )
  )
  "Mode line construct to show normal mode for ryo-modal.")
(defvar-local bard-modeline-ryo-modal-insert
  '(:eval
    (when (and (mode-line-window-selected-p)
                (bard/ryo-insert-p)
                (not (derived-mode-p 'Info-mode 'help-mode '
special-mode 'message-mode))))
      (propertize "<I>" 'face 'prot-modeline-indicator-blue-bg)
    )
  )
  "Mode line construct to show insert mode for ryo-modal.")
;;; Input method
(defvar-local prot-modeline-input-method
  '(:eval
    (when current-input-method-title
      (propertize (format " %s " current-input-method-title)
                    'face 'prot-modeline-indicator-green-bg
                    'mouse-face 'mode-line-highlight)))
  "Mode line construct to report the multilingual environment.")
;;; Buffer status
(defvar-local prot-modeline-buffer-status
  '(:eval
    (when (file-remote-p default-directory)
      (propertize " @ "
                    'face 'prot-modeline-indicator-red-bg
                    'mouse-face 'mode-line-highlight)))
  "Mode line construct for showing remote file name.")
;;; Dedicated window
(defvar-local prot-modeline-window-dedicated-status

```

```

'(:eval
  (when (window-dedicated-p)
    (propertize " = "
      'face 'prot-modeline-indicator-magenta-bg
      'mouse-face 'mode-line-highlight)))
"Mode line construct for dedicated window indicator.")
;;; Buffer name and modified status
(defun prot-modeline-buffer-identification-face ()
  "Return appropriate face or face list for `
  prot-modeline-buffer-identification'."
  (let ((file (buffer-file-name)))
    (cond
      ((and (mode-line-window-selected-p)
            file
            (buffer-modified-p))
       '(error italic mode-line-buffer-id))
      ((and file (buffer-modified-p))
       'italic)
      ((mode-line-window-selected-p)
       'mode-line-buffer-id))))
(defun prot-modeline--buffer-name ()
  "Return `buffer-name', truncating it if necessary.
  See `prot-modeline-string-truncate'."
  (when-let ((name (buffer-name)))
    (prot-modeline-string-truncate name)))
(defun prot-modeline-buffer-name ()
  "Return buffer name, with read-only indicator if relevant."
  (let ((name (prot-modeline--buffer-name)))
    (if buffer-read-only
        (format "%s %s" (char-to-string #xE0A2) name)
        name)))
(defun prot-modeline-buffer-name-help-echo ()
  "Return `help-echo' value for `prot-modeline-buffer-identification'."
  (concat
    (propertize (buffer-name) 'face 'mode-line-buffer-id)
    "\n"
    (propertize
      (or (buffer-file-name)
          (format "No underlying file.\nDirectory is: %s"
                default-directory))
      'face 'mode-line-buffer-id)))

```

```

'face 'font-lock-doc-face)))
(defvar-local prot-modeline-buffer-identification
  '(:eval
    (propertize (prot-modeline-buffer-name)
      'face (prot-modeline-buffer-identification-face)
      'mouse-face 'mode-line-highlight
      'help-echo (prot-modeline-buffer-name-help-echo)))
  "Mode line construct for identifying the buffer being displayed.
  Propertize the current buffer with the `mode-line-buffer-id'
  face. Let other buffers have no face.")
;;; Major mode
(defun prot-modeline-major-mode-indicator ()
  "Return appropriate propertized mode line indicator for the major
  mode."
  (let ((indicator (cond
    ((derived-mode-p 'text-mode) "$")
    ((derived-mode-p 'prog-mode) "λ")
    ((derived-mode-p 'term-mode) ">_")
    ((derived-mode-p 'emms-playlist-mode) "♪")
    (t "$"))))
    (propertize indicator 'face 'shadow)))
(defun prot-modeline-major-mode-name ()
  "Return capitalized `major-mode' without the -mode suffix."
  (capitalize (string-replace "-mode" "" (symbol-name major-mode))))
(defun prot-modeline-major-mode-help-echo ()
  "Return `help-echo' value for `prot-modeline-major-mode'."
  (if-let ((parent (get major-mode 'derived-mode-parent)))
    (format "Symbol: `%s'. Derived from: `%s'" major-mode parent)
    (format "Symbol: `%s'." major-mode)))
(defvar-local prot-modeline-major-mode
  (list
    (propertize "%[" 'face 'prot-modeline-indicator-red)
    '(:eval
      (concat
        (prot-modeline-major-mode-indicator)
        " "
        (propertize
          (prot-modeline-string-abbreviate-but-last
            (prot-modeline-major-mode-name)
            2)
          'mouse-face 'mode-line-highlight

```

```

        'help-echo (prot-modeline-major-mode-help-echo)))
      (propertize "%]" 'face 'prot-modeline-indicator-red))
    "Mode line construct for displaying major modes.")
  (defvar-local prot-modeline-process
    (list '(" mode-line-process))
    "Mode line construct for the running process indicator.")
  ;;;; Git branch and diffstat
  (declare-function vc-git--symbolic-ref "vc-git" (file))
  (defun prot-modeline--vc-branch-name (file backend)
    "Return capitalized VC branch name for FILE with BACKEND."
    (when-let ((rev (vc-working-revision file backend))
               (branch (or (vc-git--symbolic-ref file)
                           (substring rev 0 7))))
      (capitalize branch)))
  (declare-function vc-git-working-revision "vc-git" (file))
  (defvar prot-modeline-vc-map
    (let ((map (make-sparse-keymap)))
      (define-key map [mode-line down-mouse-1] 'vc-diff)
      (define-key map [mode-line down-mouse-3] 'vc-root-diff)
      map)
    "Keymap to display on VC indicator.")
  (defun prot-modeline--vc-help-echo (file)
    "Return `help-echo' message for FILE tracked by VC."
    (format "Revision: %s\nmouse-1: `vc-diff'\nmouse-3: `vc-root-diff'"
            (vc-working-revision file)))
  (defun prot-modeline--vc-text (file branch &optional face)
    "Prepare text for Git controlled FILE, given BRANCH.
With optional FACE, use it to propertize the BRANCH."
    (concat
      (propertize (char-to-string #xE0A0) 'face 'shadow)
      " "
      (propertize branch
                   'face face
                   'mouse-face 'mode-line-highlight
                   'help-echo (prot-modeline--vc-help-echo file)
                   'local-map prot-modeline-vc-map)
      ;; " "
      ;; (prot-modeline-diffstat file)
      ))
  (defun prot-modeline--vc-details (file branch &optional face)

```

```

    "Return Git BRANCH details for FILE, truncating it if necessary.
    The string is truncated if the width of the window is smaller
    than `split-width-threshold'."
    (prot-modeline-string-truncate
      (prot-modeline--vc-text file branch face)))
  (defvar prot-modeline--vc-faces
    '((added . vc-locally-added-state)
      (edited . vc-edited-state)
      (removed . vc-removed-state)
      (missing . vc-missing-state)
      (conflict . vc-conflict-state)
      (locked . vc-locked-state)
      (up-to-date . vc-up-to-date-state))
    "VC state faces.")
  (defun prot-modeline--vc-get-face (key)
    "Get face from KEY in `prot-modeline--vc-faces'."
    (alist-get key prot-modeline--vc-faces 'up-to-date))
  (defun prot-modeline--vc-face (file backend)
    "Return VC state face for FILE with BACKEND."
    (prot-modeline--vc-get-face (vc-state file backend)))
  (defvar-local prot-modeline-vc-branch
    '(:eval
      (when-let* (((mode-line-window-selected-p))
                  (file (buffer-file-name))
                  (backend (vc-backend file))
                  ;; ((vc-git-registered file))
                  (branch (prot-modeline--vc-branch-name file
                                                                backend)))
        (face (prot-modeline--vc-face file backend)))
      (prot-modeline--vc-details file branch face)))
    "Mode line construct to return propertized VC branch.")
  ;;;; Flymake errors, warnings, notes
  (declare-function flymake--severity "flymake" (type))
  (declare-function flymake-diagnostic-type "flymake" (diag))
  ;; Based on `flymake--mode-line-counter'.
  (defun prot-modeline-flymake-counter (type)
    "Compute number of diagnostics in buffer with TYPE's severity.
    TYPE is usually keyword `:error', `:warning' or `:note'."
    (let ((count 0))
      (dolist (d (flymake-diagnostics))
        (when (= (flymake--severity type)
                  (flymake-diagnostic-type d))
          (incf count)))))

```



```

        (flymake--severity (flymake-diagnostic-type d)))
      (cl-incf count)))
    (when (cl-plusp count)
      (number-to-string count))))
(defvar prot-modeline-flymake-map
  (let ((map (make-sparse-keymap)))
    (define-key map [mode-line down-mouse-1] '
      flymake-show-buffer-diagnostics)
    (define-key map [mode-line down-mouse-3] '
      flymake-show-project-diagnostics)
    map)
  "Keymap to display on Flymake indicator.")
(defmacro prot-modeline-flymake-type (type indicator &optional face)
  "Return function that handles Flymake TYPE with stylistic
  INDICATOR and FACE."
  `(defun ,(intern (format "prot-modeline-flymake-%s" type)) ()
    (when-let ((count (prot-modeline-flymake-counter
                      ,(intern (format ":%s" type)))))
      (concat
        (propertize ,indicator 'face 'shadow)
        (propertize count
                     'face ',(or face type)
                     'mouse-face 'mode-line-highlight
                     ;; FIXME 2023-07-03: Clicking on the text with
                     ;; this buffer and a single warning present,
the
                     ;; diagnostics take up the entire frame. Why?
                     'local-map prot-modeline-flymake-map
                     'help-echo "mouse-1: buffer diagnostics\
nmouse-3: project diagnostics")))))
(prot-modeline-flymake-type error "✖")
(prot-modeline-flymake-type warning "!")
(prot-modeline-flymake-type note "•" success)
(defvar-local prot-modeline-flymake
  `(:eval
    (when (and (bound-and-true-p flymake-mode)
               (mode-line-window-selected-p))
      (list
        ;; See the calls to the macro `prot-modeline-flymake-type'
        '(:eval (prot-modeline-flymake-error))
        '(:eval (prot-modeline-flymake-warning))

```

```

        '(:eval (prot-modeline-flymake-note))))))
"Mode line construct displaying `flymake-mode-line-format'.
Specific to the current window's mode line.")
(with-eval-after-load 'eglot
  (setq mode-line-misc-info
    (delete '(eglot--managed-mode (" [" eglot--mode-line-format
    "]" "))) mode-line-misc-info)))
(defvar-local prot-modeline-eglot
  `(:eval
    (when (and (featurep 'eglot) (mode-line-window-selected-p))
      '(eglot--managed-mode eglot--mode-line-format)))
  "Mode line construct displaying Eglot information.
Specific to the current window's mode line.")
(defvar-local prot-modeline-notmuch-indicator
  '(notmuch-indicator-mode
    (" "
      (:eval (when (mode-line-window-selected-p)
                  notmuch-indicator--counters)))))
"The equivalent of `notmuch-indicator-mode-line-construct'.
Display the indicator only on the focused window's mode line.")
(defvar-local bard-evil-state-indicator
  '(:eval
    (when (and (bound-and-true-p evil-local-mode)
                (mode-line-window-selected-p))
      (let ((state-label
              (pcase evil-state
                ('normal (propertize " <N>" 'face '
prot-modeline-indicator-green))
                ('insert (propertize " <I>" 'face '
prot-modeline-indicator-blue))
                ('visual (propertize " <V>" 'face '
prot-modeline-indicator-yellow))
                ('replace (propertize " <R>" 'face '
prot-modeline-indicator-red))
                ('emacs (propertize " <E>" 'face '
prot-modeline-indicator-magenta))
                ('motion (propertize " <V>" 'face '
prot-modeline-indicator-cyan))
                (_ (propertize " <>" 'face 'shadow)))))
        state-label)))
"Modeline indicator for current Evil state.")

```

```

;;; Org clock indicator
(defun bard-modeline--org-clock-check-overflow ()
  "Check if the clock is overflow."
  (when org-clock-effort
    (> (org-clock-get-clocked-time)
        (org-duration-to-minutes org-clock-effort))))
(defun bard-modeline--org-clock-string ()
  "Make a string for org-clock indicator."
  (let* ((clocked-mins (org-clock-get-clocked-time))
        (work-time (org-duration-from-minutes clocked-mins)))
    (if org-clock-effort
        (format "[%s/%s]"
                  work-time
                  org-clock-effort)
        (format "[%s]" work-time))))
(defun bard-modeline--org-clock-choose-face ()
  "Make the background red when overflowing the clock."
  (if (bard-modeline--org-clock-check-overflow)
      'prot-modeline-indicator-red-bg
      'prot-modeline-indicator-green-bg))
(defvar-local bard-modeline-org-clock
  '(:eval
    (when (and (featurep 'org)
                (bound-and-true-p org-clock-current-task)
                (mode-line-window-selected-p))
      (propertize
        (concat " ⌚ "
                (prot-modeline-string-truncate
                 (bard-modeline--org-clock-string))
                " ")
        'face (bard-modeline--org-clock-choose-face)
        'mouse-face 'mode-line-highlight
        'help-echo "Org clocked task\nmouse-1: Go to task"
        'local-map (let ((map (make-sparse-keymap)))
                      (define-key map [mode-line down-mouse-1]
                        #'org-clock-goto)
                      map))))))
  "Mode line construct showing the current Org clocked task.")
;;; Miscellaneous
;; (setq global-mode-string
;;       (remove 'org-mode-line-string global-mode-string))

```

```

;; (delq 'org-mode-line-string global-mode-string)
(with-eval-after-load 'org
  (setq mode-line-misc-info
    (delete 'org-mode-line-string mode-line-misc-info)))
(defvar-local prot-modeline-misc-info
  '(:eval
    (when (mode-line-window-selected-p)
      mode-line-misc-info))
  "Mode line construct displaying `mode-line-misc-info'.
Specific to the current window's mode line.")
(defvar-local prot-modeline-frame-name
  '(:eval
    (when-let* ((_ (mode-line-window-selected-p))
                (current-frame (selected-frame))
                (_ (frame-live-p current-frame))
                (parameters (frame-parameters))
                (name (capitalize (alist-get 'name parameters))))
      (format "%s %s "
        "@ "
        (propertize name
                     'face 'prot-modeline-indicator-blue
                     'mouse-face 'mode-line-highlight
                     'help-echo (format "Current frame name\
nmouse-1: `buffer-menu'")))))
  "Mode line construct to display the current frame name.")
;;; Risky local variables
;; NOTE 2023-04-28: The `risky-local-variable' is critical, as those
;; variables will not work without it.
(dolist (construct '(prot-modeline-kbd-macro
  prot-modeline-narrow
  bard-modeline-centered-cursor
  bard-evil-state-indicator
  prot-modeline-input-method
  bard-modeline-org-clock
  prot-modeline-buffer-status
  prot-modeline-window-dedicated-status
  prot-modeline-frame-name
  prot-modeline-evil
  prot-modeline-buffer-identification
  prot-modeline-major-mode
  prot-modeline-process

```

```

        prot-modeline-vc-branch
        prot-modeline-flymake
        prot-modeline-eglot
        prot-modeline-misc-info
        prot-modeline-notmuch-indicator))
  (put construct 'risky-local-variable t))
  (provide 'bard-modeline)

```

3.8 bard-package

I renamed the functions from [this post](#) because otherwise I would forget I had them in my system. I keep personal/custom functions prefixed with **bard/** so I can look at them through minibuffer if I forgot the name. This shows me that they are not part of default Emacs.

```

;; taken and renamed functions from
(defun bard/package-report ()
  "Report total package counts grouped by archive."
  (interactive)
  (package-refresh-contents)
  (bard/display-package-report
   (let* ((arch-pkgs (bard/archive-packages))
          (counts (seq-sort-by #'cdr #'> (bard/archive-counts
arch-pkgs)))
          (by-arch (seq-group-by #'car arch-pkgs)))
     (concat
      (format "Total packages: %s\n\n" (apply #'+ (mapcar #'cdr
counts)))
      (mapconcat
       (lambda (archive)
        (concat "• "
                (format "%s (%s)" (car archive) (cdr archive))
                ": "
                (mapconcat (lambda (ap-pair) (cdr ap-pair))
                           (alist-get (car archive) by-arch)
                           ", "))))
       counts
       "\n\n")))))
(defun bard/display-package-report (output)
  "Display OUTPUT in a popup buffer."
  (let ((buffer-name "*package-report*"))
    (with-help-window buffer-name

```

```

    (with-current-buffer buffer-name
      (visual-line-mode 1)
      (erase-buffer)
      (insert output)
      (goto-char (point-min))))))
(defun bard/archive-packages ()
  "Return a list of (archive . package) cons cells."
  (seq-reduce
    (lambda (res package)
      (let ((archive (package-desc-archive
                      (cadr (assq package package-archive-contents))))
            )
        (pkg (symbol-name package)))
      (push (cons archive pkg) res)))
    (mapcar #'car package-alist)
    nil))
(defun bard/archive-counts (arch-pkgs)
  "Return a list of cons cells from alist ARCH-PKGS.
The cars are package archives, the cdrs are the number of
packages installed from each archive."
  (seq-reduce
    (lambda (counts key)
      (cons (cons key (+ 1 (or (cdr (assoc key counts)) 0)))
            (assoc-delete-all key counts)))
    (mapcar #'car arch-pkgs)
    nil))
(provide 'bard-package)

```

3.9 bard-theme

3.9.1 Fontaine font presets

The main configuration for fontaine is found at **bard-emacs-theme** - Fonts. These are the presets for different fonts that I can switch between using **C-c f**.

```

(setq fontaine-presets
  '((default
    :default-height 140
    :default-family "Iosevka Comfy"
    :variable-pitch-family "Iosevka Comfy"
    :variable-pitch-height 1.0

```

```

:fixed-pitch-family "Iosevka Comfy"
:fixed-pitch-height 1.0
:bold-weight bold
)
(tiny
:inherit default
:default-height 135)
(wide
:default-height 135
:default-family "Iosevka Comfy Wide"
:fixed-pitch-family "Iosevka Comfy Wide"
:variable-pitch-family "Iosevka Comfy Wide Motion Duo")
(prot
:default-family "Iosevka Comfy Wide Motion"
:default-height 140
:default-weight medium
:fixed-pitch-family "Iosevka Comfy Wide Motion"
:variable-pitch-family "Iosevka Comfy Wide Motion Duo"
:bold-weight extrabold)
(duo
:default-family "Iosevka Comfy Motion"
:default-height 140
:default-weight medium
:fixed-pitch-family "Iosevka Comfy Motion"
:variable-pitch-family "Iosevka Comfy Motion Duo"
:bold-weight extrabold)
(mono
:default-height 140
:default-family "Terminus"
:fixed-pitch-family "Terminus"
:variable-pitch-family "Terminus")
(mono-large
:inherit mono
:default-height 150
:variable-pitch-height 160)
(mac
:default-height 130
:default-family "Monaco"
:variable-pitch-family "Monaco"
:fixed-pitch-family "Monaco")
(large

```

```

      :inherit default
      :default-height 160)
(huge
  :inherit default
  :default-height 180)
(t
  :default-family "Monospace"))))

```

3.9.2 Japanese CJK font

Explanation provided by [AJATT — Japanese fonts](#). Essentially: Chinese fonts and Japanese fonts look different for certain Kanji. Since I don't want Chinese glyphs prioritized in Emacs over Japanese ones, I set the font to the Japanese (JP) version of Noto CJK font which is a very accessible and popular font for Asian languages.

```

(set-fontset-font
 t 'han
 (font-spec :family "Noto Serif CJK JP") nil 'prepend)

```

3.9.3 Switching themes

```

(defun bard/disable-all-themes ()
  "disable all active themes."
  (interactive)
  (dolist (i custom-enabled-themes)
    (disable-theme i)))
(defvar bard/after-theme-load-hook nil
  "Hook that runs after a new theme is loaded using `bard/
  select-theme`."
  )
(dolist (hook '(bard/after-theme-load-hook))
  (add-hook hook #'logos-update-fringe-in-buffers))
(defun bard/select-theme (&optional theme)
  "Enable the specified THEME, or prompt the user to select one if
  THEME is nil."
  (interactive
   (list
    (completing-read "Select theme: "
                      (mapcar 'symbol-name (custom-available-themes))
                      nil nil))))
(let* ((theme-symbol (if (symbolp theme) theme (intern theme)))
       (theme-name (symbol-name theme-symbol)))

```



```

      (display-theme-name (if (string-suffix-p "-theme"
theme-name)
                             (substring theme-name 0 -6)
                             theme-name))
      (colored-theme-name (propertize display-theme-name 'face
'(:weight bold))))
      (bard/disable-all-themes)
      (load-theme theme-symbol t)
      (message "Loaded the %s theme" colored-theme-name)
      (run-hooks 'bard/after-theme-load-hook))

```

3.9.4 Updating cursor for text mode

```

(defvar my-last-cursor-type nil)
(defun bard/update-cursor-type ()
  "Set cursor type to 'bar in text modes, 'box otherwise.
leave it alone in pdf-view-mode."
  (unless (derived-mode-p 'pdf-view-mode)
    (let ((new-cursor (if (derived-mode-p 'text-mode) 'bar 'box)))
      (unless (eq my-last-cursor-type new-cursor)
        (setq cursor-type new-cursor)
        (setq my-last-cursor-type new-cursor)))))
(add-hook 'post-command-hook #'bard/update-cursor-type)

```

3.9.5 Heading faces

```

(defun bard/outline-heading-faces ()
  (set-face-attribute 'org-document-title nil
                      :inherit '(outline-1 variable-pitch)
                      :weight 'light
                      :height 1.5)
  (set-face-attribute 'org-level-1 nil
                      :inherit '(outline-2 variable-pitch)
                      :weight 'light
                      :height 1.3)
  (set-face-attribute 'org-level-2 nil
                      :inherit 'outline-3
                      :height 1.2)
  (set-face-attribute 'org-level-3 nil
                      :inherit '(outline-4 variable-pitch)
                      :height 1.1)

```

```
(set-face-attribute 'org-level-4 nil
                   :inherit '(outline-5 variable-pitch)
                   :height 1.1)
(set-face-attribute 'org-level-5 nil
                   :inherit '(outline-6 variable-pitch)
                   :height 1.1)
(set-face-attribute 'org-level-6 nil
                   :inherit '(outline-6 variable-pitch)
                   :height 1.1)
(set-face-attribute 'org-agenda-date nil
                   :inherit 'variable-pitch
                   :weight 'bold
                   :height 1.3)
(set-face-attribute 'org-agenda-structure nil
                   :inherit 'variable-pitch
                   :weight 'bold
                   :height 1.5))
(add-hook 'bard/after-theme-load-hook #'bard/outline-heading-faces)
```

3.9.6 Provide library

```
(provide 'bard-theme)
```

3.10 bard-web

3.10.1 Load required libraries

```
(require 'emms)
(require 'elfeed-search)
```

3.10.2 Elfeed/EMMS YouTube code

I made a video about this here: <https://bardman.dev/technology/elfeed>.

```
(defun bard/play-elfeed-video ()
  "Play the URL of the entry at point in mpv if it's a YouTube video
  ."
  (interactive)
  (let ((entry (elfeed-search-selected :single)))
    (if entry
        (let ((url (elfeed-entry-link entry)))
```

```

        (if (and url (string-match-p "https?://\\(www\\.\\.\\)?
youtube\\.com\\|youtu\\.be" url))
            (progn
                (async-shell-command (format "mpv '%s'" url))
                (elfeed-search-untag-all-unread))
            (message "The URL is not a YouTube link: %s" url)))
        (message "No entry selected in Elfeed."))))
(defun bard/add-video-emms-queue ()
  "Play the URL of the entry at point in mpv if it's a YouTube video
  . Add it to EMMS queue."
  (interactive)
  (let ((entry (elfeed-search-selected :single)))
    (if entry
        (let ((url (elfeed-entry-link entry)))
          (if (and url (string-match-p "https?://\\(www\\.\\.\\)?
youtube\\.com\\|youtu\\.be" url))
              (let* ((playlist-name "Watch Later")
                     (playlist-buffer (get-buffer (format " *%s*"
playlist-name))))
                (unless playlist-buffer
                  (setq playlist-buffer (emms-playlist-new (format "
%s*" playlist-name))))
                (emms-playlist-set-playlist-buffer playlist-buffer)
                (emms-add-url url)
                (elfeed-search-untag-all-unread)
                (message "Added YouTube video to EMMS playlist: %s"
url))
              (message "The URL is not a YouTube link: %s" url)))
        (message "No entry selected in Elfeed."))))
(defun bard/add-video-watch-later ()
  "Add the current Elfeed YouTube entry URL to '~/Videos/watch-later
.m3u' and mark it as read."
  (interactive)
  (let ((entry (elfeed-search-selected :single)))
    (if entry
        (let* ((url (elfeed-entry-link entry))
               (watch-later-file (expand-file-name "~/Videos/
watch-later.m3u")))
          (if (and url (string-match-p "https?://\\(www\\.\\.\\)?
youtube\\.com\\|youtu\\.be" url))
              (progn

```

```

(with-temp-buffer
  (insert (concat url "\n"))
  (append-to-file (point-min) (point-max)
    watch-later-file))
;; Remove the 'unread tag from the entry directly
(setf (elfeed-entry-tags entry)
  (remove 'unread (elfeed-entry-tags entry)))
;; Force UI update
(when (derived-mode-p 'elfeed-search-mode)
  (elfeed-search-update-entry entry))
(message "Added video to watch later: %s" url))
(message "The URL is not a YouTube link: %s" url)))
(message "No entry selected in Elfeed.")))

```

3.10.3 Provide library

```
(provide 'bard-web)
```

3.11 bard-window

This module has some code from [Protesilaos's dotemacs](#), especially the [prot-window.el](#) file. I copied most of this code a long time ago, so his library may have changed since then. But for me it works, and I never had to toy with it. No reason to reinvent the wheel with this one.

```

(require 'prot-common)
(defvar prot-window-window-sizes
  '( :max-height (lambda () (floor (frame-height) 3))
    :min-height 10
    :max-width (lambda () (floor (frame-width) 4))
    :min-width 20)
  "Property list of maximum and minimum window sizes.
The property keys are `:max-height', `:min-height', `:max-width',
and `:min-width'. They all accept a value of either a
number (integer or floating point) or a function.")
(defun prot-window--get-window-size (key)
  "Extract the value of KEY from `prot-window-window-sizes'."
  (when-let ((value (plist-get prot-window-window-sizes key)))
    (cond
      ((functionp value)
       (funcall value))
      ((numberp value)
       value)

```

```

      (t
        (error "The value of '%s' is neither a number nor a function"
          key))))))
(defun prot-window-select-fit-size (window)
  "Select WINDOW and resize it.
The resize pertains to the maximum and minimum values for height
and width, per `prot-window-window-sizes'.
Use this as the `body-function' in a `display-buffer-alist' entry."
  (select-window window)
  (fit-window-to-buffer
    window
    (prot-window--get-window-size :max-height)
    (prot-window--get-window-size :min-height)
    (prot-window--get-window-size :max-width)
    (prot-window--get-window-size :min-width))
  ;; If we did not use `display-buffer-below-selected', then we must
  ;; be in a lateral window, which has more space. Then we do not
  ;; want to dedicate the window to this buffer, because we will be
  ;; running out of space.
  (when (or (window-in-direction 'above) (window-in-direction 'below))
    (set-window-dedicated-p window t)))
(defun prot-window--get-display-buffer-below-or-pop ()
  "Return list of functions for `
prot-window-display-buffer-below-or-pop'."
  (list
    #'display-buffer-reuse-mode-window
    (if (or (prot-common-window-small-p)
      (prot-common-three-or-more-windows-p))
      #'display-buffer-below-selected
      #'display-buffer-pop-up-window)))
(defun prot-window-display-buffer-below-or-pop (&rest args)
  "Display buffer below current window or pop a new window.
The criterion for choosing to display the buffer below the
current one is a non-nil return value for
`prot-common-window-small-p'.
Apply ARGS expected by the underlying `display-buffer' functions.
This as the action function in a `display-buffer-alist' entry."
  (let ((functions (prot-window--get-display-buffer-below-or-pop)))
    (catch 'success
      (dolist (fn functions)

```

```

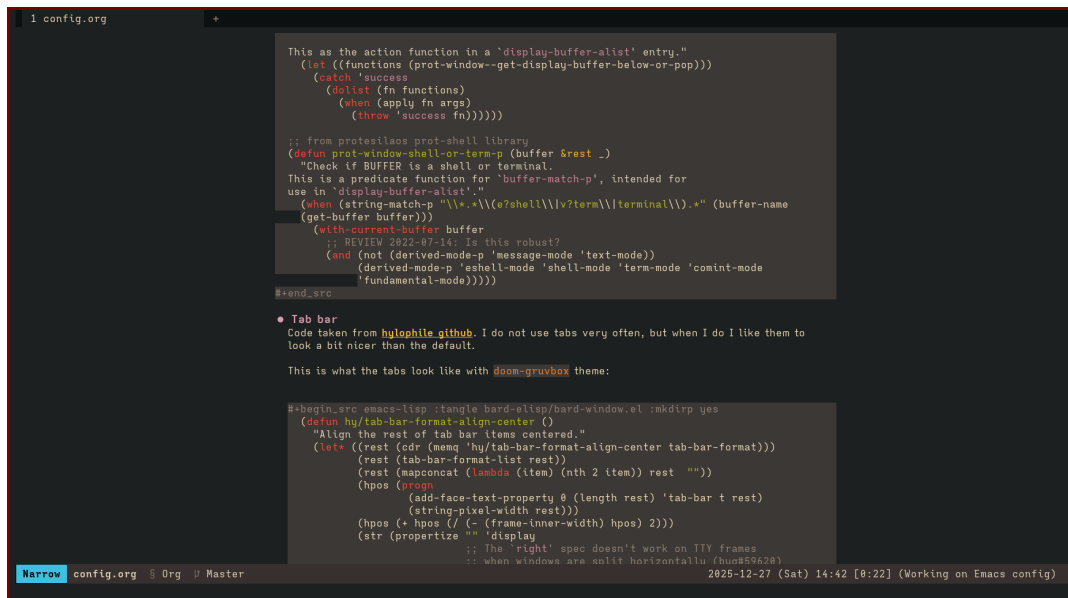
      (when (apply fn args)
        (throw 'success fn))))))
;; from protesilaos prot-shell library
(defun prot-window-shell-or-term-p (buffer &rest _)
  "Check if BUFFER is a shell or terminal.
This is a predicate function for `buffer-match-p', intended for
use in `display-buffer-alist'."
  (when (string-match-p "\\*..*\\((e?shell\\|v?term\\|terminal\\)).*" (
    buffer-name (get-buffer buffer)))
    (with-current-buffer buffer
      ;; REVIEW 2022-07-14: Is this robust?
      (and (not (derived-mode-p 'message-mode 'text-mode))
        (derived-mode-p 'eshell-mode 'shell-mode 'term-mode '
          comint-mode 'fundamental-mode))))))

```

3.11.1 Tab bar

Code taken from [hylophile github](#). I do not use tabs very often, but when I do I like them to look a bit nicer than the default.

This is what the tabs look like with `doom-gruvbox` theme:



```

(defun hy/tab-bar-format-align-center ()
  "Align the rest of tab bar items centered."
  (let* ((rest (cdr (memq 'hy/tab-bar-format-align-center
    tab-bar-format))))

```

```

        (rest (tab-bar-format-list rest))
        (rest (mapconcat (lambda (item) (nth 2 item)) rest ""))
        (hpos (progn
                (add-face-text-property 0 (length rest) 'tab-bar t
rest)
                (string-pixel-width rest)))
        (hpos (+ hpos (/ (- (frame-inner-width) hpos) 2)))
        (str (propertize "" 'display
                        ;; The `right' spec doesn't work on TTY
frames
                        ;; when windows are split horizontally (
bug#59620)
                        (if (window-system)
                            `(space :align-to (- right (,hpos)))
                            `(space :align-to (,(- (
frame-inner-width) hpos))))))
        `((align-center menu-item ,str ignore))))
(setq tab-bar-tab-name-format-function #'hy/
      tab-bar-tab-name-format-default)
(defun hy/tab-bar-tab-name-format-default (tab i)
  (let* ((hint (format "%d" i))
         (name (alist-get 'name tab))
         (dir (concat "(" (alist-get 'dir tab "") " ")))
         (name-format (concat
                        " "
                        (propertize hint 'face 'tab-bar-hint)
                        " "
                        name
                        " ")))
    (add-face-text-property
      0 (length name-format)
      (funcall tab-bar-tab-face-function tab)
      'append name-format)
    name-format))
(setq tab-bar-tab-name-function #'hy/tab-bar-tab-name-current)
(defun hy/tab-bar-tab-name-current ()
  (hy/shorten-string
   (hy/abbreviate-tab-name
    (buffer-name (window-buffer (or (minibuffer-selected-window)
                                     (and (window-minibuffer-p)
                                           (get-mru-window)))))))

```

```

25))
(defun hy/set-tab-dir ()
  (setf (alist-get 'dir (cdr (tab-bar--current-tab-find)))
        (hy/tab-bar-dir)))
(defun hy/abbreviate-directory-path (path)
  "Turns `~/code/test/t` into `~/c/t/project`."
  (let* ((directories (seq-filter (lambda (s) (not (string= s "")))
                                (split-string path "/"))))
    (last-dir (car (last directories)))
    (abbreviated-dirs (mapcar (lambda (dir)
                              (if (string= dir last-dir)
                                  dir
                                  (substring dir 0 (if (
string-prefix-p "." dir) 2 1))))
                              directories)))
    (mapconcat 'identity abbreviated-dirs "/")))
(defun hy/tab-bar-dir ()
  (hy/shorten-string (hy/abbreviate-directory-path
                    (abbreviate-file-name
                     (or (projectile-project-root)
                         default-directory)))
                    10
                    t))
(defun hy/shorten-string (string max-length &optional at-start)
  (let ((len (length string)))
    (if (> len max-length)
        (if at-start
            (concat "... " (substring string (- len max-length) len))
            (concat (substring string 0 max-length) "..."))
        string)))
(defun hy/abbreviate-tab-name (name)
  (string-trim (replace-regexp-in-string
                (rx (or "*" "helpful" "Org Src"))
                "" name)))

```

3.11.2 Toggle window split direction

```

(defun bard/toggle-window-split ()
  "Toggle between horizontal and vertical window splits, preserving
  buffer layout."
  (interactive)

```



```

(let ((current-buffers (mapcar #'window-buffer (window-list))) ;
      List of buffers in current windows
      (split-direction (if (= (window-width) (frame-width))
                            'vertical
                            'horizontal)))
  (delete-other-windows)
  ;; Toggle the split direction
  (if (eq split-direction 'horizontal)
      (split-window-vertically)
      (split-window-horizontally))
  ;; Restore buffers to the new windows
  (let ((windows (window-list)))
    (cl-loop for buffer in current-buffers
              for window in windows
              do (set-window-buffer window buffer))))
(provide 'bard-window)

```

3.12 bard-writing

3.12.1 Load required libraries

```

(require 'consult)
(require 'beframe)
(require 'calendar)
(require 'org-roam-node)
(require 'denote)

```

3.12.2 Note buffers

```

(defvar bard/consult--source-notes
  `(:name      "Note Buffers"
     :narrow    ?n
     :category  buffer
     :face      consult-buffer
     :history   buffer-name-history
     :items     ,(lambda ()
                   (mapcar #'buffer-name
                           (seq-filter
                            (lambda (buf)
                              (string-prefix-p "[Note]" (
buffer-name buf))))
                           (beframe-buffer-list))))

```

```

        :action      ,#'switch-to-buffer
        :state       ,#'consult--buffer-state)
"Consult source for note buffers (limited to beframe buffers).")
(defun bard/consult-buffer-notes ()
  "Show `consult-buffer` limited to buffers starting with [Note]."
  (interactive)
  (consult-buffer '(bard/consult--source-notes)))
(defun bard/ibuffer-notes ()
  "Open `ibuffer` limited to buffers starting with [Note]."
  (interactive)
  (ibuffer nil "*Ibuffer-Notes*"
    '((name . "^\\[Note\\]"))))

```

3.12.3 Searching notes

```

(use-package consult-denote
  :ensure t
  :bind
  ("C-c n g" . consult-denote-grep)
  ("C-c n f" . consult-denote-find)
  :config
  (setq consult-denote-find-command 'consult-fd
        consult-denote-grep-command 'consult-ripgrep))

```

3.12.4 University Class jumping in uni.org file

```

(defvar bard/class-dirs
  '(("COMM 111" . "~/Documents/Uni/SUMMER2026-COMM 111/")
    ("CSE 220" . "~/Documents/Uni/SUMMER2026-CSE 220/")
    ("PHYS 296" . "~/Documents/Uni/SUMMER2026-PHYS 296/")
    ("PHYS 299" . "~/Documents/Uni/SUMMER2026-PHYS 299/"))
  "Mapping of class names to their document directories.")
(defvar bard/uni-notes-file "~/Notes/denote/uni.org"
  "Path to the main university org file.")
(defun bard/jump-to-class (class &optional with-dired)
  "Jump to CLASS heading in bard/uni-notes-file, can open its dir in
  Dired."
  (interactive)
  (list
   (completing-read "Class: " (mapcar #'car bard/class-dirs))
   current-prefix-arg))

```

```

(let ((dir (cdr (assoc class bard/class-dirs))))
  (delete-other-windows)
  (find-file bard/uni-notes-file)
  (widen)
  (goto-char (point-min))
  (search-forward class nil nil)
  (when with-dired
    (let ((dired-window (split-window-right)))
      (with-selected-window dired-window
        (dired dir))))))
(defun bard/jump-to-class-new-frame (class &optional with-dired)
  "Open CLASS notes and dir in a new frame titled after CLASS,
  optionally WITH-DIRED."
  (interactive
   (list
    (completing-read "Class: " (mapcar #'car bard/class-dirs)
      current-prefix-arg)))
  (let* ((dir (cdr (assoc class bard/class-dirs)))
        (frame (make-frame `((frame-title-format . ,class)))))
    (select-frame-set-input-focus frame)
    (delete-other-windows)
    (find-file bard/uni-notes-file)
    (widen)
    (goto-char (point-min))
    (search-forward class nil nil)
    (when with-dired
      (let ((dired-window (split-window-right)))
        (with-selected-window dired-window
          (dired dir)
          (bframe-rename-current-frame))))))
;; Optional: bind to a key
(global-set-key (kbd "C-c u") #'bard/jump-to-class)
(global-set-key (kbd "C-c U") #'bard/jump-to-class-new-frame)

```

```

(defvar bard/subheading-keys
  '(("schedule" . "s")
    ("check in" . "c")
    ("homework" . "h")
    ("notes" . "n")
    ("exams" . "e")
    ("todos" . "t")))

```

```

"Fixed key bindings for known class subheadings in the transient
menu.")
(defun bard/get-class-subheadings (class)
  "Return a list of level-2 subheading titles under CLASS in `bard/
  uni-notes-file'."
  (with-current-buffer (find-file-noselect bard/uni-notes-file)
    (save-excursion
      (widen)
      (goto-char (point-min))
      (search-forward class nil nil)
      (let ((class-end (save-excursion
                          (org-end-of-subtree t)
                          (point))))
        (subheadings '()))
      (while (re-search-forward "^\\*\\* \\(\\.+\\)" class-end t)
        (push (match-string-no-properties 1) subheadings))
      (nreverse subheadings))))
(defun bard/jump-to-class-subheading (class subheading &optional
  with-dired)
  "Jump to SUBHEADING under CLASS in `bard/uni-notes-file'."
  (let ((dir (cdr (assoc class bard/class-dirs))))
    (delete-other-windows)
    (find-file bard/uni-notes-file)
    (widen)
    (goto-char (point-min))
    (search-forward class nil nil)
    (let ((class-pos (point)))
      (if (re-search-forward
          (concat "^\\*\\* " (regexp-quote subheading)) nil t)
        (progn
          (org-fold-show-entry)
          (org-narrow-to-subtree))
        (goto-char class-pos)))
      (when with-dired
        (let ((dired-window (split-window-right)))
          (with-selected-window dired-window
            (dired dir))))))
(defun bard/jump-to-class-section ()
  "Pick a class then a subheading, both via completing-read."
  (interactive)
  (let* ((class (completing-read "Class: " (mapcar #'car bard/

```

```

    class-dirs) nil t))
      (sections (bard/get-class-subheadings class))
      (section (completing-read "Section: " sections nil t))
      (dired-p current-prefix-arg))
    (bard/jump-to-class-subheading class section dired-p)))
(defvar bard/--current-class nil
  "Class selected in the first step of bard/class-menu.")
(transient-define-prefix bard/--section-menu ()
  "Pick a subheading within `bard/--current-class'."
  [:description
   (lambda () (format "Sections in %s" bard/--current-class))
   :class transient-column
   :setup-children
   (lambda (_)
     (let ((sections (bard/get-class-subheadings bard/--
current-class)))
       (delq nil
        (mapcar
         (lambda (section)
           (let ((key (cdr (assoc section bard/subheading-keys)
)))
             (when key
               (transient-parse-suffix
                'bard/--section-menu
                `(',key ,section (lambda () (interactive)
                                   (bard/
jump-to-class-subheading
                                   bard/--current-class ,
section))))))))
         sections))))))])
;; TODO make some kind of dired menu after selecting the class
;; (transient-define-prefix bard/--dired-menu ()
;;   "Pick an action relating to dired."
;;   [:description
;;    (lambda () (format "Open dired in" bard/--current-class))
;;    :class transient-column
;;    :setup-children
;;    (lambda
;;      )])
(transient-define-prefix bard/class-menu ()
  "Jump to a class section via transient menus."

```

```

["Classes"
 :class transient-column
 :setup-children
 (lambda (_)
  (let ((i 1))
   (mapcar
    (lambda (entry)
     (let* ((class (car entry))
            (key   (number-to-string i)))
      (setq i (1+ i))
      (transient-parse-suffix
       'bard/class-menu
       `(:,key ,class (lambda () (interactive)
                           (setq bard/--current-class ,class)
                           (bard/--section-menu))))))
     bard/class-dirs))))))

```

3.12.5 Typst lecture notes with denote file naming scheme

```

(defun denote-typst-with-signature ()
  "Create a Typst note while prompting for template and signature."
  (declare (interactive-only t))
  (interactive)
  (let ((denote-file-type 'typst)
        (denote-prompts
         (denote-add-prompts '(signature template))))
    (call-interactively #'denote)))

```

3.12.6 Denote journal

I want to have a custom format for my daily journal file to start what I want to get done that day specifically. It helps me get focused on what needs to be accomplished on days when I have a lot of tasks.

```

(defun bard/denote-todo-template ()
  "Return string for daily tasks heading in `denote-journal' entries
  ."
  (with-temp-buffer
   (org-mode)
   (insert (format "* Tasks for %s\n** Время я потратил
бездельничая\n\n* Notes for today\n\n* Clocktable\n"
                   (format-time-string "%Y-%m-%d (%a)"))))

```

```
(let ((org-clock-clocktable-default-properties
      '(:scope file :maxlevel 3 :link nil :compact t)))
  (org-clock-report))
(buffer-string)))
```

3.12.7 Math input mode

```
;; Taken from: https://stackoverflow.com/a/75314192
(defun add-multiple-into-list (lst items)
  "Add each item from ITEMS into LST."
  (dolist (item items)
    (add-to-list lst item)))
(defun bard/cdlatex-add-math-symbols ()
  "Add functions into list."
  (add-multiple-into-list
   'cdlatex-math-symbol-alist-comb
   '((?V "\\vec")
     (?s "\\sigma" "\\text{ s.t. }" "\\sin")
     (?= "\\implies" "\\Leftrightarrow" "\\Longlefttrightarrow"))))
;; (use-package xenops
;;   :ensure t
;;   :config
;;   (setq xenops-math-image-scale-factor 1.2))
(define-minor-mode bard/org-math-mode
  "Enable features to write math in `org-mode'."
  :init-value nil
  :lighter " S="
  :global nil
  (xenops-mode t)
  (org-cdlatex-mode t)
  (electric-pair-local-mode t)
  (bard/cdlatex-add-math-symbols)
  ;; sending math to calc from latex doc
  (define-key org-mode-map (kbd "C-S-e") #'latex-math-from-calc))
;; latex into calc
(defun latex-math-from-calc ()
  "Evaluate `calc' on the contents of line at point."
  (interactive)
  (cond ((region-active-p)
        (let* ((beg (region-beginning))
               (end (region-end))
```

```

        (string (buffer-substring-no-properties beg end)))
      (kill-region beg end)
      (insert (calc-eval `,(string calc-language latex
                                   calc-prefer-frac t
                                   calc-angle-mode rad))))))
    (t (let ((l (thing-at-point 'line)))
        (end-of-line 1) (kill-line 0)
        (insert (calc-eval `,(l
                                calc-language latex
                                calc-prefer-frac t
                                calc-angle-mode rad))))))))

```

3.12.8 Provide library

```
(provide 'bard-writing)
```

3.13 prot-common

This library is mostly used in the `bard-modeline.el` library. It is mostly unmodified from the time when I copied it from [Protesilaos's Emacs configuration](#) over 2 years ago.

```

;;; prot-common.el --- Common functions for my dotemacs -*-
lexical-binding: t -*-
;; Copyright (C) 2020-2023 Protesilaos Stavrrou
;; Author: Protesilaos Stavrrou <info@protesilaos.com>
;; URL: https://protesilaos.com/emacs/dotemacs
;; Version: 0.1.0
;; Package-Requires: ((emacs "30.1"))
;; This file is NOT part of GNU Emacs.
;; This program is free software; you can redistribute it and/or
modify
;; it under the terms of the GNU General Public License as published
by
;; the Free Software Foundation, either version 3 of the License, or
(at
;; your option) any later version.
;;
;; This program is distributed in the hope that it will be useful,
;; but WITHOUT ANY WARRANTY; without even the implied warranty of
;; MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
;; GNU General Public License for more details.
;;

```



```

;; You should have received a copy of the GNU General Public License
;; along with this program.  If not, see <https://www.gnu.org/licenses/>.
;;; Commentary:
;;
;; Common functions for my Emacs: <https://protesilaos.com/emacs/dotemacs/>.
;;
;; Remember that every piece of Elisp that I write is for my own
;; educational and recreational purposes.  I am not a programmer and
;; I
;; do not recommend that you copy any of this if you are not certain
;; of
;; what it does.
;;; Code:
(eval-when-compile
  (require 'subr-x)
  (require 'cl-lib))
(defgroup prot-common ()
  "Auxiliary functions for my dotemacs."
  :group 'editing)
;;;###autoload
(defun prot-common-number-even-p (n)
  "Test if N is an even number."
  (if (numberp n)
      (= (% n 2) 0)
      (error "%s is not a number" n)))
;;;###autoload
(defun prot-common-number-integer-p (n)
  "Test if N is an integer."
  (if (integerp n)
      n
      (error "%s is not an integer" n)))
;;;###autoload
(defun prot-common-number-integer-positive-p (n)
  "Test if N is a positive integer."
  (if (prot-common-number-integer-p n)
      (> n 0)
      (error "%s is not a positive integer" n)))
;; Thanks to Gabriel for providing a cleaner version of
;; `prot-common-number-negative': <https://github.com/gabriel376>.

```

```

;;;###autoload
(defun prot-common-number-negative (n)
  "Make N negative."
  (if (and (numberp n) (> n 0))
      (* -1 n)
      (error "%s is not a valid positive number" n)))
;;;###autoload
(defun prot-common-reverse-percentage (number percent change-p)
  "Determine the original value of NUMBER given PERCENT.
CHANGE-P should specify the increase or decrease. For simplicity,
nil means decrease while non-nil stands for an increase.
NUMBER must satisfy `numberp', while PERCENT must be `natnump'."
  (unless (numberp number)
    (user-error "NUMBER must satisfy numberp"))
  (unless (natnump percent)
    (user-error "PERCENT must satisfy natnump"))
  (let* ((pc (/ (float percent) 100))
         (pc-change (if change-p (+ 1 pc) pc))
         (n (if change-p pc-change (float (- 1 pc-change)))))
    ;; FIXME 2021-12-21: If float, round to 4 decimal points.
    (/ number n)))
;;;###autoload
(defun prot-common-percentage-change (n-original n-final)
  "Find percentage change between N-ORIGINAL and N-FINAL numbers.
When the percentage is not an integer, it is rounded to 4
floating points: 16.666666666666664 => 16.667."
  (unless (numberp n-original)
    (user-error "N-ORIGINAL must satisfy numberp"))
  (unless (numberp n-final)
    (user-error "N-FINAL must satisfy numberp"))
  (let* ((difference (float (abs (- n-original n-final))))
         (n (* (/ difference n-original) 100))
         (round (floor n)))
    ;; FIXME 2021-12-21: Any way to avoid the `string-to-number'?
    (if (> n round) (string-to-number (format "%0.4f" n)) round)))
;; REVIEW 2023-04-07 07:43 +0300: I just wrote the conversions from
;; seconds. Hopefully they are correct, but I need to double check.
(defun prot-common-seconds-to-minutes (seconds)
  "Convert a number representing SECONDS to MM:SS notation."
  (let ((minutes (/ seconds 60))
        (seconds (% seconds 60)))

```

```

    (format "%.2d:%.2d" minutes seconds)))
(defun prot-common-seconds-to-hours (seconds)
  "Convert a number representing SECONDS to HH:MM:SS notation."
  (let* ((hours (/ seconds 3600))
         (minutes (/ (% seconds 3600) 60))
         (seconds (% seconds 60)))
    (format "%.2d:%.2d:%.2d" hours minutes seconds)))
;;;###autoload
(defun prot-common-seconds-to-minutes-or-hours (seconds)
  "Convert SECONDS to either minutes or hours, depending on the
value."
  (if (> seconds 3599)
      (prot-common-seconds-to-hours seconds)
      (prot-common-seconds-to-minutes seconds)))
;;;###autoload
(defun prot-common-rotate-list-of-symbol (symbol)
  "Rotate list value of SYMBOL by moving its car to the end.
Return the first element before performing the rotation.
This means that if `sample-list' has an initial value of `(one
two three)', this function will first return `one' and update the
value of `sample-list' to `(two three one)'. Subsequent calls
will continue rotating accordingly."
  (unless (symbolp symbol)
    (user-error "%s is not a symbol" symbol))
  (when-let* ((value (symbol-value symbol))
              (list (and (listp value) value))
              (first (car list)))
    (set symbol (append (cdr list) (list first)))
    first))
;;;###autoload
(defun prot-common-empty-buffer-p ()
  "Test whether the buffer is empty."
  (or (= (point-min) (point-max))
      (save-excursion
        (goto-char (point-min))
        (while (and (looking-at "^\\([a-zA-Z]+: ?\\)?$")
                    (zerop (forward-line 1)))
          (eobp))))))
;;;###autoload
(defun prot-common-minor-modes-active ()
  "Return list of active minor modes for the current buffer."

```

```

(let ((active-modes))
  (mapc (lambda (m)
          (when (and (boundp m) (symbol-value m))
            (push m active-modes)))
        minor-mode-list)
    active-modes))
;;;###autoload
(defun prot-common-truncate-lines-silently ()
  "Toggle line truncation without printing messages."
  (let ((inhibit-message t))
    (toggle-truncate-lines t)))
;; NOTE 2023-08-12: I tried the `clear-message-function', but it did
;; not work. What I need is very simple and this gets the job done.
;;;###autoload
(defun prot-common-clear-minibuffer-message (&rest _)
  "Print an empty message to clear the echo area.
Use this as advice :after a noisy function."
  (message ""))
;;;###autoload
(defun prot-common-disable-hl-line ()
  "Disable Hl-Line-Mode (for hooks)."
  (hl-line-mode -1))
;;;###autoload
(defun prot-common-window-bounds ()
  "Return start and end points in the window as a cons cell."
  (cons (window-start) (window-end)))
;;;###autoload
(defun prot-common-page-p ()
  "Return non-nil if there is a `page-delimiter' in the buffer."
  (or (save-excursion (re-search-forward page-delimiter nil t))
      (save-excursion (re-search-backward page-delimiter nil t))))
;;;###autoload
(defun prot-common-window-small-p ()
  "Return non-nil if window is small.
Check if the `window-width' or `window-height' is less than
`split-width-threshold' and `split-height-threshold',
respectively."
  (or (and (numberp split-width-threshold)
          (< (window-total-width) split-width-threshold))
      (and (numberp split-height-threshold)
          (> (window-total-height) split-height-threshold))))

```

```

;;;###autoload
(defun prot-common-three-or-more-windows-p (&optional frame)
  "Return non-nil if three or more windows occupy FRAME.
If FRAME is non-nil, inspect the current frame."
  (>= (length (window-list frame :no-minibuffer)) 3))
;;;###autoload
(defun prot-common-read-data (file)
  "Read Elisp data from FILE."
  (with-temp-buffer
    (insert-file-contents file)
    (read (current-buffer))))
;;;###autoload
(defun prot-common-completion-category ()
  "Return completion category."
  (when-let ((window (active-minibuffer-window)))
    (with-current-buffer (window-buffer window)
      (completion-metadata-get
        (completion-metadata (buffer-substring-no-properties
                              (minibuffer-prompt-end)
                              (max (minibuffer-prompt-end) (point))))
        minibuffer-completion-table
        minibuffer-completion-predicate)
      'category))))
;; Thanks to Omar íAntoln Camarena for providing this snippet!
;;;###autoload
(defun prot-common-completion-table (category candidates)
  "Pass appropriate metadata CATEGORY to completion CANDIDATES.
This is intended for bespoke functions that need to pass
completion metadata that can then be parsed by other
tools (e.g. `embark')."
  (lambda (string pred action)
    (if (eq action 'metadata)
        `(metadata (category . ,category))
        (complete-with-action action candidates string pred))))
;;;###autoload
(defun prot-common-completion-table-no-sort (category candidates)
  "Pass appropriate metadata CATEGORY to completion CANDIDATES.
Like `prot-common-completion-table' but also disable sorting."
  (lambda (string pred action)
    (if (eq action 'metadata)
        `(metadata (category . ,category))

```

```

        (display-sort-function . ,#'identity))
      (complete-with-action action candidates string pred))))
;; Thanks to Igor Lima for the `prot-common-crm-exclude-selected-p':
;; <https://github.com/0x462e41>.
;; This is used as a filter predicate in the relevant prompts.
(defvar crm-separator)
;;;###autoload
(defun prot-common-crm-exclude-selected-p (input)
  "Filter out INPUT from `completing-read-multiple'.
Hide non-destructively the selected entries from the completion
table, thus avoiding the risk of inputting the same match twice.
To be used as the PREDICATE of `completing-read-multiple'."
  (if-let* ((pos (string-match-p crm-separator input))
            (rev-input (reverse input))
            (element (reverse
                      (substring rev-input 0
                                (string-match-p crm-separator
                                                  rev-input)))))
    (flag t))
  (progn
    (while pos
      (if (string= (substring input 0 pos) element)
        (setq pos nil)
        (setq input (substring input (1+ pos))
              pos (string-match-p crm-separator input)
              flag (when pos t))))
    (not flag))
  t))
;; The `prot-common-line-regexp-p' and `
prot-common--line-regexp-alist'
;; are contributed by Gabriel: <https://github.com/gabriel376>.
They
;; provide a more elegant approach to using a macro, as shown
further
;; below.
(defvar prot-common--line-regexp-alist
  '((empty . "[\s\t]*$")
    (indent . "^[ \t]+")
    (non-empty . "^.+$")
    (list . "\\\\[ \t#*+]+\\\\[0-9]+[^\s]?[.]+\\\\")
    (heading . "[=-]+"))

```

```

    "Alist of regexp types used by `prot-common-line-regexp-p'.")
(defun prot-common-line-regexp-p (type &optional n)
  "Test for TYPE on line.
TYPE is the car of a cons cell in
`prot-common--line-regexp-alist'. It matches a regular
expression.
With optional N, search in the Nth line from point."
  (save-excursion
    (goto-char (line-beginning-position))
    (and (not (bobp))
         (or (beginning-of-line n) t)
         (save-match-data
          (looking-at
           (alist-get type prot-common--line-regexp-alist))))))
;; The `prot-common-shell-command-with-exit-code-and-output'
  function is
;; courtesy of Harold Carr, who also sent a patch that improved
;; `prot-eww-download-html' (from the `prot-eww.el' library).
;;
;; More about Harold: <http://haroldcarr.com/about/>.
(defun prot-common-shell-command-with-exit-code-and-output (command
  &rest args)
  "Run COMMAND with ARGS.
Return the exit code and output in a list."
  (with-temp-buffer
    (list (apply 'call-process command nil (current-buffer) nil args
      )
          (buffer-string))))
(defvar prot-common-url-regexp
  (concat
    "~?\\<\\([-a-zA-Z0-9+&@#/%?=_|!:,.;]*\\)"
    "[.@"
    "\\([-a-zA-Z0-9+&@#/%?=_|!:,.;]+\\)\\>/?")
  "Regular expression to match (most?) URLs or email addresses.")
(autoload 'auth-source-search "auth-source")
;;;###autoload
(defun prot-common-auth-get-field (host prop)
  "Find PROP in `auth-sources' for HOST entry."
  (when-let ((source (auth-source-search :host host)))
    (if (eq prop :secret)
        (funcall (plist-get (car source) prop))

```

```

        (plist-get (flatten-list source) prop))))
;;;###autoload
(defun prot-common-parse-file-as-list (file)
  "Return the contents of FILE as a list of strings.
Strings are split at newline characters and are then trimmed for
negative space.
Use this function to provide a list of candidates for
completion (per `completing-read')."
  (split-string
   (with-temp-buffer
     (insert-file-contents file)
     (buffer-substring-no-properties (point-min) (point-max)))
   "\n" :omit-nulls "[\s\f\t\n\r\v]+"))
(defun prot-common-ignore (&rest _)
  "Use this as override advice to make a function do nothing."
  nil)
;; NOTE 2023-06-02: The `prot-common-wcag-formula' and
;; `prot-common-contrast' are taken verbatim from my `modus-themes'
;; and renamed to have the prefix `prot-common-' instead of
;; `modus-themes-'. This is all my code, of course, but I do it
  this
;; way to ensure that this file is self-contained in case someone
;; copies it.
;; This is the WCAG formula: <https://www.w3.org/TR/WCAG20-TECHS/G18.html>.
(defun prot-common-wcag-formula (hex)
  "Get WCAG value of color value HEX.
The value is defined in hexadecimal RGB notation, such #123456."
  (cl-loop for k in '(0.2126 0.7152 0.0722)
    for x in (color-name-to-rgb hex)
    sum (* k (if (<= x 0.03928)
      (/ x 12.92)
      (expt (/ (+ x 0.055) 1.055) 2.4)))))
;;;###autoload
(defun prot-common-contrast (c1 c2)
  "Measure WCAG contrast ratio between C1 and C2.
C1 and C2 are color values written in hexadecimal RGB."
  (let ((ct (/ (+ (prot-common-wcag-formula c1) 0.05)
    (+ (prot-common-wcag-formula c2) 0.05))))
    (max ct (/ ct))))
;;; EXPERIMENTAL macros (not meant to be used anywhere)

```



```

;; TODO 2023-09-30: Try the same with `cl-defmacro' and &key
(defmacro prot-common-if (condition &rest consequences)
  "Separate the CONSEQUENCES of CONDITION semantically.
Like `if', `when', `unless' but done by using `:then' and `:else'
keywords. The forms under each keyword of `:then' and `:else'
belong to the given subset of CONSEQUENCES.
- The absence of `:else' means: (if CONDITION (progn CONSEQUENCES)).
- The absence of `:then' means: (if CONDITION nil CONSEQUENCES).
- Otherwise: (if CONDITION (progn then-CONSEQUENCES)
  else-CONSEQUENCES)."
  (declare (indent 1))
  (let (then-consequences else-consequences last-kw)
    (dolist (elt consequences)
      (let ((is-keyword (keywordp elt)))
        (cond
         ((and (not is-keyword) (eq last-kw :then))
          (push elt then-consequences))
         ((and (not is-keyword) (eq last-kw :else))
          (push elt else-consequences))
         ((and is-keyword (eq elt :then))
          (setq last-kw :then))
         ((and is-keyword (eq elt :else))
          (setq last-kw :else))))))
    `(if ,condition
      ,(if then-consequences
        `(progn ,@(nreverse then-consequences))
        nil)
      ,@(nreverse else-consequences))))
(provide 'prot-common)
;;; prot-common.el ends here

```